Analytical Chemistry

Topic 16

Key ideas

Unit 53 Qualitative analysis — detecting the presence of inorganic chemical species

- Identifying cations in a sample
- Identifying common gases and vapours
- Identifying anions in a sample
- Devising a scheme to separate a mixture of species
- Risks associated with chemical tests

Unit 54 Tests for functional groups, separation and purification of compounds

- Chemical tests for functional groups in carbon compounds
- Distillation and fractional distillation
- Liquid-liquid extraction
- Re-crystallization
- Chromatography
- Checking the purity of a product

Unit 55 Quantitative methods of analysis

- Gravimetric analysis **Extension**
- Precipitation titrations
- Redox titrations involving potassium permanganate
- Redox titrations involving iodine

Unit 56 Instrumental analytical methods

- Colorimetry
- Infrared spectroscopy
- Mass spectrometry

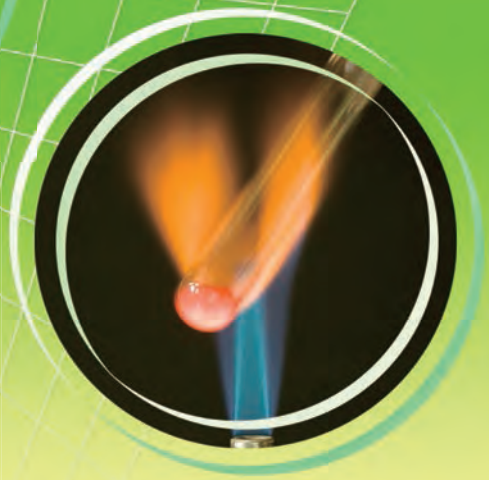
Unit 57 Contribution of analytical chemistry to our society

- Food and drug analyses
- Air quality control
- Forensic chemistry
- Clinical laboratory tests









Unit 53 Qualitative analysis — detecting the presence of inorganic chemical species

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

53.1 Introduction to analytical chemistry

Analytical chemistry plays an important role in nearly all aspects of chemistry, for example, agricultural, clinical, environmental, forensic, manufacturing and pharmaceutical chemistry. The nitrogen content of a fertilizer determines its value. Foods must be analyzed for contaminants (e.g. pesticide residues) and for essential nutrients (e.g. vitamin content). The air in cities must be analyzed for pollutants.

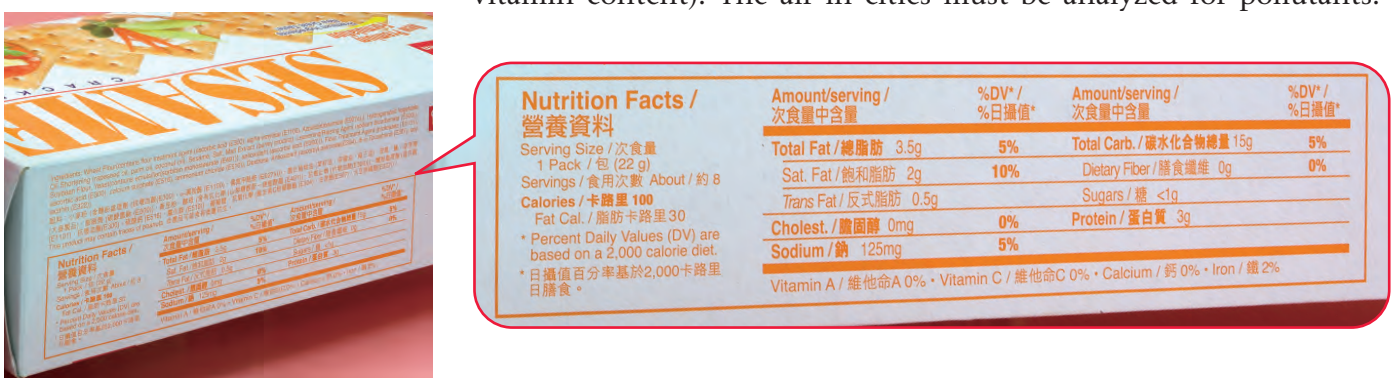


Fig. 53.1 What nutrients are present in this box of biscuits?

The discipline of analytical chemistry consists of **qualitative** and **quantitative** analyses. The former deals with the identification of chemical species present in a sample, while the latter deals with the determination of how much a constituent is present.

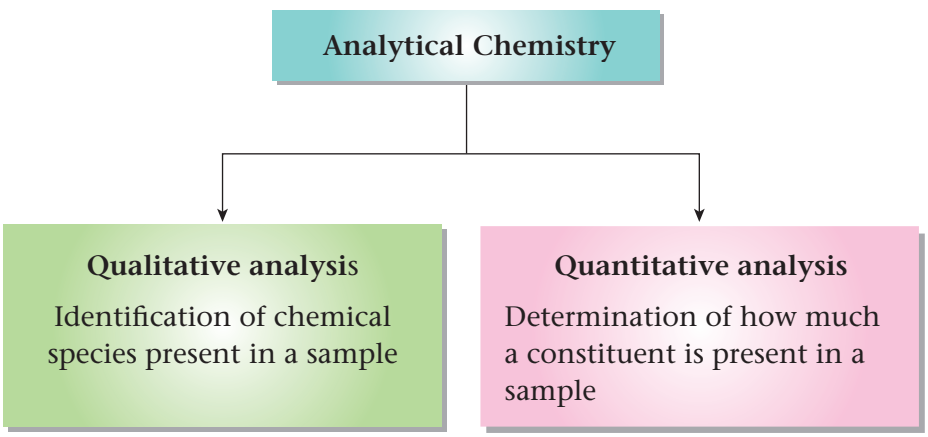


Fig. 53.2 The discipline of analytical chemistry consists of qualitative and quantitative analyses

qualitative 定性　　quantitative 定量

We can perform qualitative tests by simple chemical reactions. For example, the formation of a white precipitate (silver chloride) when adding dilute nitric acid followed by silver nitrate solution to a sample solution indicates the presence of chloride ions in the sample.

$$Ag^+(aq) + Cl^-(aq) \longrightarrow \underset{\text{silver chloride}}{AgCl(s)}$$

We will discuss the tests for inorganic chemical species in this unit, followed by tests for functional groups in carbon compounds in the next one. In this unit, we will study chemical tests for identifying cations and anions in samples.

53.2 Identifying the cation in a sample

We shall use a number of cations to illustrate the general approach for identifying the cation in a sample:

- aluminium ions, Al^{3+};
- ammonium ions, NH_4^+;
- calcium ions, Ca^{2+};
- copper(II) ions, Cu^{2+};
- iron(II) ions, Fe^{2+};
- iron(III) ions, Fe^{3+};
- lead(II) ions, Pb^{2+};
- magnesium ions, Mg^{2+};
- potassium ions, K^+;
- sodium ions, Na^+; and
- zinc ions, Zn^{2+}.

We will look at each of the following tests in turn:

- appearance;
- solubility in water;
- flame test;
- action of heat;
- action of dilute hydrochloric acid;

- action of dilute sulphuric acid;
- action of dilute aqueous solution of sodium hydroxide; and
- action of dilute aqueous ammonia.

53.3 Appearance

The colour of a solid sample may give a clue to its identity or the cation it contains (Table 53.1). However, there are several possibilities for a given colour. Therefore we cannot take the colour of a sample as a positive identification of the identity of the sample or the cation it contains. Further chemical tests are required.

Table 53.1 Deductions based on the colour of solids

Colour of solid	Deduction
Yellow / reddish yellow	• lead(II) oxide, PbO; or • silver iodide, AgI; or • hydrated iron(III) chloride, $FeCl_3$
Green	• hydrated iron(II) salts (pale green); or • hydrated copper(II) chloride, $CuCl_2$; or • copper(II) carbonate, $CuCO_3$ — CE 2003 Paper 2 Q29
Blue	• hydrated copper(II) salts

Yellow / reddish yellow



lead(II) oxide

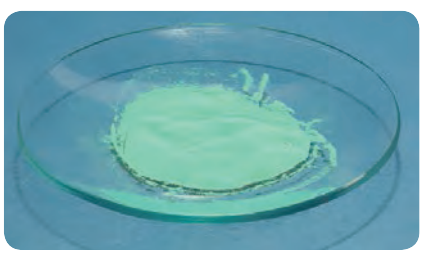


hydrated iron(III) chloride

Green



hydrated copper(II) chloride (bluish green)



copper(II) carbonate

Fig. 53.3 Some coloured solids mentioned in Table 53.1

The colour of an aqueous solution of a sample may give a clue to the cation it contains (Table 53.2).

Table 53.2 Deductions based on the colour of aqueous solutions of samples

Colour of aqueous solution	Deduction
Yellow-brown	iron(III) ions, Fe^{3+}
Pale green	iron(II) ions, Fe^{2+}
Blue or green	copper(II) ions, Cu^{2+}

CE 1999 Paper 2 Q13
CE 2001 Paper 2 Q1
CE 2003 Paper 2 Q29

53.4 Solubility in water

* We have discussed the solubility of common ionic compounds in water in Topic 4 Acids and Bases.

Knowledge of the solubility of ionic compounds in water is very useful in the identification of ions in samples. Table 53.3 shows the solubility of some common ionic compounds in water*.

Notice that all nitrates, and all common salts of sodium, potassium and ammonium are soluble in water.

Table 53.3 Solubility of some ionic compounds in water

Ionic compounds	Solubility in water
Nitrates	all are soluble
Halides (chlorides, bromides and iodides)	all are soluble except: silver halides, lead(II) halides (ASL 2012 Paper 1 Q9; AL 2012 Paper 1 Q7(c))
Carbonates and sulphites	all are insoluble except: sodium carbonate and sodium sulphite; potassium carbonate and potassium sulphite; ammonium carbonate
Sulphates	all are soluble except: barium sulphate, calcium sulphate (CE 2011 Paper 1 Q1(b)), lead(II) sulphate

53.5 Flame test

* We have described the procedure for carrying out flame test in Topic 1 Planet Earth.

When compounds of some metals are heated in a flame, they give distinctive flame colours*. Fig. 53.4 shows some examples. Therefore we can use the flame colours to identify particular cations present in samples.

Video
Flame test

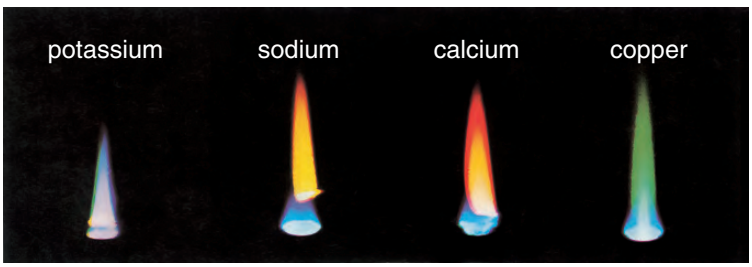


Fig. 53.4 Flame colours of compounds of some metals

Table 53.4 Characteristic flame colours of some cations

Flame colour	Cation in sample
Lilac	potassium ions, K^+
Golden yellow	sodium ions, Na^+
Brick-red	calcium ions, Ca^{2+}
Bluish green	copper(II) ions, Cu^{2+}

ASL 2001 Q7
CE 2005 Paper 1 Q12
CE 2011 Paper 1 Q1(b)
AL 1999 Paper 1 Q7(c)(ii)
CE 2011 Paper 2 Q40
AL 2002 Paper 2 Q1(c)(i)
CE 2005 Paper 2 Q5

Sometimes two cations give similar flame colours. Therefore we should not use the flame test as the only identification method.

53.6 Action of heat

Most solids undergo changes when heated. These changes often provide a clue to the cation and / or anion present in the solid. For example, changes in a solid sample during heating and the colour of the solid formed provide clues to the cation in the sample. The identity of the gas given off, if any, provides a clue to the anion in the sample.

Now we will look at deductions concerning cations in samples based on observations made during heating (Table 53.5). We will discuss the deductions concerning anions based on observations made during heating later in this unit.

During the test, place a little of the solid sample in a dry ignition tube and heat gently at first and then strongly until no further changes occur.

ASL 2012 Paper 1 Q9
AL 2012 Paper 1 Q7(c)

Table 53.5 Deductions concerning the cation in a sample based on observations made during heating

Observation		Deduction	
White sublimate*		ammonium halides may be present N1	
Colour of solid formed		**Solid formed is**	**Cation in sample**
Yellow	yellow when hot; white when cold	zinc oxide	zinc
Orange	orange when hot; yellow when cold	lead(II) oxide	lead
Reddish brown	reddish brown when hot; brown when cold	iron(III) oxide	iron
Black	black solid from blue or green substance	copper(II) oxide	copper

* When ammonium chloride is heated, it seems to sublime. In fact, it decomposes into ammonia gas and hydrogen chloride gas on heating. The products recombine to form the original compound on cooling.

$$NH_4Cl(s) \underset{\text{cool}}{\overset{\text{heat}}{\rightleftharpoons}} NH_3(g) + HCl(g)$$



Fig. 53.5 White sublimate of ammonium halide forms upon heating and deposits on the upper, cooler part of the tube



(a) yellow when hot



(b) white when cold

Fig. 53.6 Zinc oxide

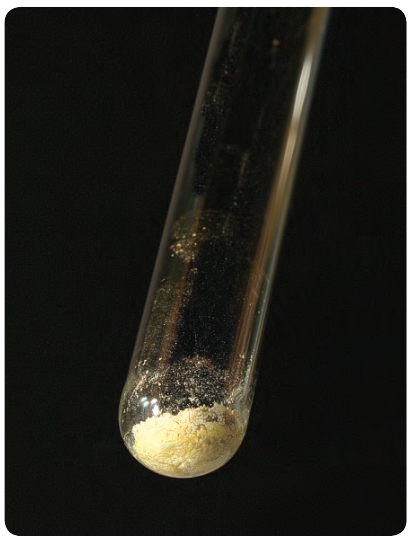
N1 Separating a mixture of $NH_4Cl(s)$, $PbCl_2(s)$ and $MgCl_2(s)$

Separate $NH_4Cl(s)$, $PbCl_2(s)$ and $MgCl_2(s)$ from a mixture of the three compounds in the following way:

- Heat the mixture. Only $NH_4Cl(s)$ sublimes. It can be collected on a cold surface.
- Add water to the remaining solid mixture.
- $PbCl_2(s)$ is insoluble. It can be collected by filtration.
- $MgCl_2(s)$ can be obtained from its solution by crystallization.

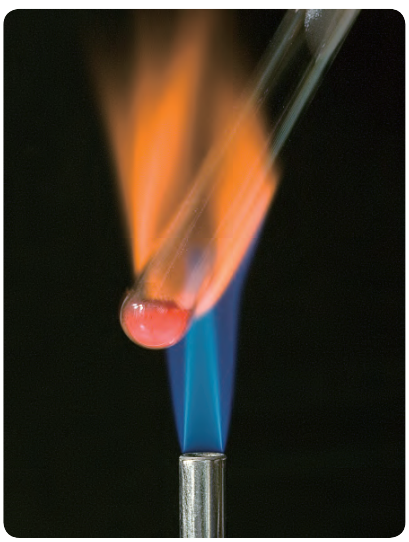


(a) orange when hot



(b) yellow when cold

Fig. 53.7 Lead(II) oxide



(a) reddish brown when hot



(b) brown when cold

Fig. 53.8 Iron(III) oxide

53.7 Action of dilute hydrochloric acid

The chloride ions in dilute hydrochloric acid may form insoluble metal chlorides with cations in solutions.

During the test, prepare a solution if the sample is soluble in water. Then add dilute hydrochloric acid to the sample solution.

AL 2011 Paper 1 Q7(b)(i)

A sample solution containing lead(II) ions or silver ions gives a white precipitate (Fig. 53.9). Both ions form insoluble, white chlorides.

$$Pb^{2+}(aq) + 2Cl^{-}(aq) \longrightarrow PbCl_2(s)$$

$$Ag^{+}(aq) + Cl^{-}(aq) \longrightarrow AgCl(s)$$

The lead(II) chloride precipitate dissolves in hot water to give a colourless solution. The silver chloride precipitate is soluble in aqueous ammonia.

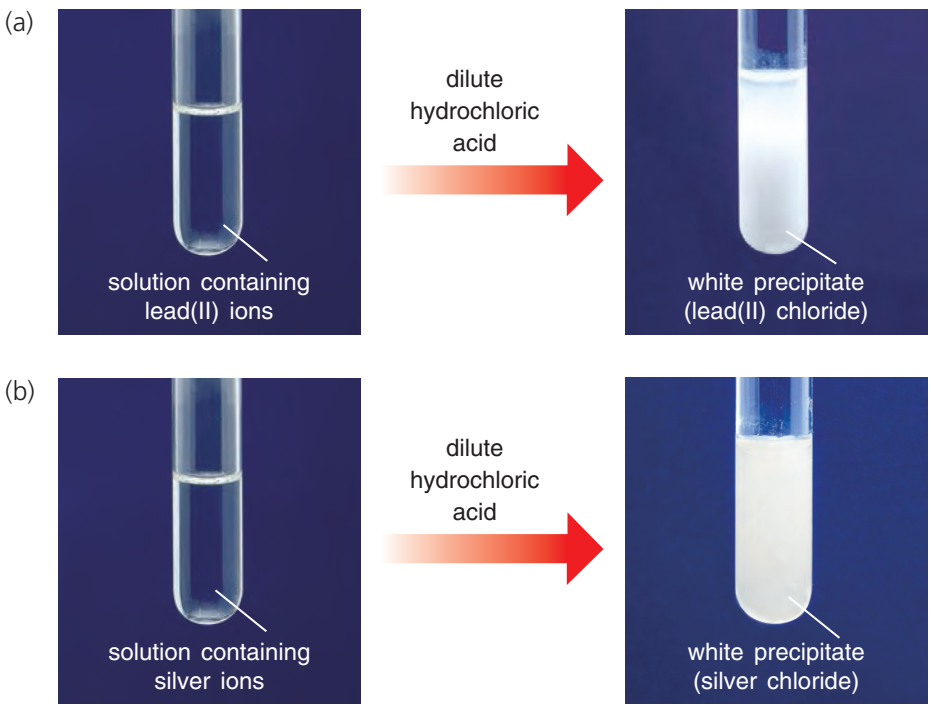


Fig. 53.9 The formation of lead(II) chloride and silver chloride precipitates

53.8 Action of dilute sulphuric acid

The sulphate ions in dilute sulphuric acid may form insoluble metal sulphates with cations in solutions.

During the test, prepare a solution if the sample is soluble in water. Then add dilute sulphuric acid to the sample solution.

A sample solution containing calcium ions or lead(II) ions gives a white precipitate. Both ions form insoluble sulphates.

$$Ca^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow CaSO_4(s)$$

$$Pb^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow PbSO_4(s)$$

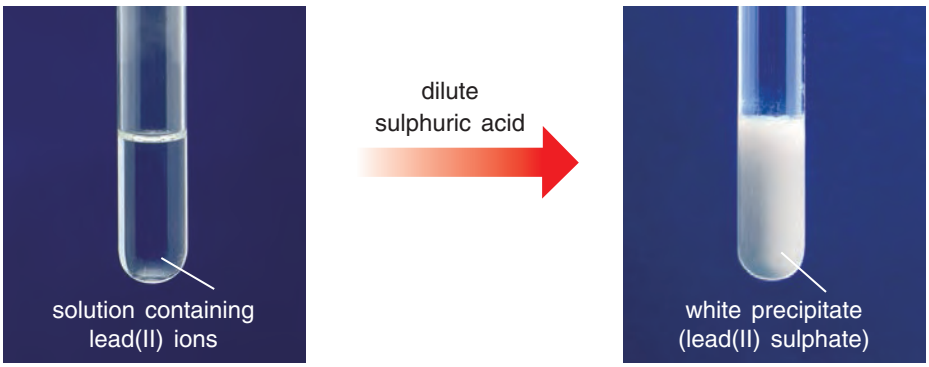


Fig. 53.10 The formation of lead(II) sulphate precipitate



53.1 Deciding whether a precipitate will form when mixing each pair of solutions

Consider the following pairs of solutions:

a) $Ba(NO_3)_2(aq) + MgSO_4(aq)$

b) $CuSO_4(aq) + Na_2CO_3(aq)$

i) For each pair, decide whether the solutions will produce a precipitate when mixed.

ii) Write a chemical equation and an ionic equation for each case in which a precipitate is produced.

Solution

a) A white precipitate (barium sulphate) will be produced.

$Ba(NO_3)_2(aq) + MgSO_4(aq) \longrightarrow BaSO_4(s) + Mg(NO_3)_2(aq)$

$Ba^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow BaSO_4(s)$

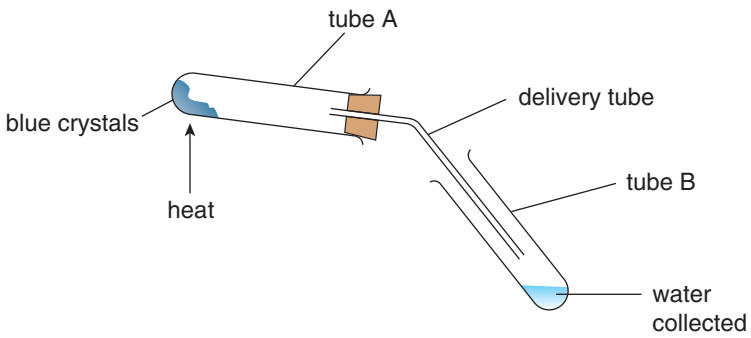
b) A bluish green precipitate (copper(II) carbonate) will be produced.

$CuSO_4(aq) + Na_2CO_3(aq) \longrightarrow CuCO_3(s) + Na_2SO_4(aq)$

$Cu^{2+}(aq) + CO_3^{2-}(aq) \longrightarrow CuCO_3(s)$

53.2 Heating some hydrated crystals

The set-up shown below was used to collect the liquid water given off when some hydrated blue crystals were heated strongly for some time.



a) Explain why tube A was clamped in a slanting position as shown.

b) Why was it desirable to keep the free end of the delivery tube above the liquid collected in tube B?

c) After strong heating, a black residue was left in tube A. Suggest an ion that might be present in the crystals based on this observation. Explain your answer.

Solution

a) To prevent any water condensed in the cool part of the tube from running back to the hot part and cracking the hot glass.

b) To prevent 'sucking back' of the water.

c) The black residue might be copper(II) oxide. Therefore copper(II) ions might be present in the crystals.

Checkpoint

Refer to page T21.

1 Consider the following pairs of solutions:

a) $CaCl_2(aq) + Na_2SO_4(aq)$] CE 2004 Paper 2 Q8

b) $ZnSO_4(aq) + NaNO_3(aq)$

c) $Pb(NO_3)_2(aq) + MgCl_2(aq)$

i) For each pair, decide whether the solutions will produce a precipitate when mixed.

ii) Write a chemical equation and an ionic equation for each case in which a precipitate is produced.

2 A compound X gave a lilac flame in flame test. When an aqueous solution of X was mixed with copper(II) sulphate solution, a precipitate appeared.

a) Suggest the cation present in X.

b) Deduce whether X contains chloride ions, nitrate ions or carbonate ions. Justify your answer.

53.9 Action of dilute aqueous solution of sodium hydroxide

* We have discussed these reactions in Topic 4 Acids and Bases.

* An amphoteric compound can behave as an acid or a base.

We can use dilute aqueous solution of sodium hydroxide to precipitate insoluble hydroxides from solutions containing some cations*. The hydroxides of some cations are amphoteric* and they dissolve in excess alkali. Table 53.6 summarizes the observations when we add dilute aqueous solution of sodium hydroxide to solutions containing some cations.

CE 2005 Paper 2 Q26
CE 2007 Paper 2 Q6

A solution containing ammonium ions, potassium ions or sodium ions does not give a precipitate with dilute aqueous solution of sodium hydroxide. A solution containing ammonium ions gives off ammonia gas when warmed with the alkali. N2

CE 2004 Paper 1 Q2(b)
CE 2005 Paper 1 Q12
CE 2006 Paper 1 Q4
CE 2008 Paper 2 Q3
AL 2001 Paper 2 Q3(d)

$$NH_4^+(aq) + OH^-(aq) \longrightarrow NH_3(g) + H_2O(l)$$

CE 2000 Paper 2 Q29
CE 2010 Paper 2 Q20

Ammonia gas turns moist red litmus paper blue.

During the test, prepare a solution if the sample is soluble in water. Then add dilute aqueous solution of sodium hydroxide to the sample solution. We can deduce the cation present in the sample based on the observations. N3



Fig. 53.11 Copper(II) hydroxide, iron(II) hydroxide, iron(III) hydroxide (left to right)

N2 Examination questions often ask about identifying ammonium compounds by warming with dilute aqueous solution of sodium hydroxide.

N3 Refer to page T22.

N4 Calcium hydroxide are precipitated only from concentrated solutions of calcium ions.

Table 53.6 Summary of observations when dilute aqueous solution of sodium hydroxide is added to solutions containing some cations

Adding NaOH(aq) to solution containing	Colour of precipitate formed	Precipitate soluble in excess NaOH(aq)?	Ionic equations(s)
Calcium ions, Ca^{2+}	white N4	✘	$Ca^{2+}(aq) + 2OH^-(aq) \longrightarrow Ca(OH)_2(s)$
Magnesium ions, Mg^{2+}	white	✘	$Mg^{2+}(aq) + 2OH^-(aq) \longrightarrow Mg(OH)_2(s)$
Copper(II) ions, Cu^{2+}	pale blue	✘	$Cu^{2+}(aq) + 2OH^-(aq) \longrightarrow Cu(OH)_2(s)$
Iron(II) ions, Fe^{2+}	green*	✘	$Fe^{2+}(aq) + 2OH^-(aq) \longrightarrow Fe(OH)_2(s)$
Iron(III) ions, Fe^{3+}	reddish brown	✘	$Fe^{3+}(aq) + 3OH^-(aq) \longrightarrow Fe(OH)_3(s)$
Aluminium ions, Al^{3+}	white	✔ (a colourless solution forms)	$Al^{3+}(aq) + 3OH^-(aq) \longrightarrow Al(OH)_3(s)$ $Al(OH)_3(s) + OH^-(aq) \longrightarrow [Al(OH)_4]^-(aq)$
Lead(II) ions, Pb^{2+}	white	✔ (a colourless solution forms)	$Pb^{2+}(aq) + 2OH^-(aq) \longrightarrow Pb(OH)_2(s)$ $Pb(OH)_2(s) + 2OH^-(aq) \longrightarrow [Pb(OH)_4]^{2-}(aq)$
Zinc ions, Zn^{2+}	white	✔ (a colourless solution forms)	$Zn^{2+}(aq) + 2OH^-(aq) \longrightarrow Zn(OH)_2(s)$ $Zn(OH)_2(s) + 2OH^-(aq) \longrightarrow [Zn(OH)_4]^{2-}(aq)$

CE 2006 Paper 2 Q7
CE 2010 Paper 2 Q20
CE 2006 Paper 1 Q4
ASL 2000 Q1(a)
CE 2008 Paper 2 Q3
CE 2007 Paper 2 Q6
AL 2006 Paper 1 Q7(c)(ii)
AL 2011 Paper 1 Q7(b)(i)
ASL 2011 Paper 2 Q4(b)
AL 2011 Paper 2 Q6(c)

N5

* The green iron(II) hydroxide turns brown on prolonged standing in air due to the formation of iron(III) hydroxide.

$4Fe(OH)_2(s) + 2H_2O(l) + O_2(g) \longrightarrow 4Fe(OH)_3(s)$

N5 A sample solution containing nickel(II) ions gives a green precipitate with dilute aqueous solution of sodium hydroxide.
$Ni^{2+}(aq) + 2OH^-(aq) \longrightarrow Ni(OH)_2(s)$
green precipitate

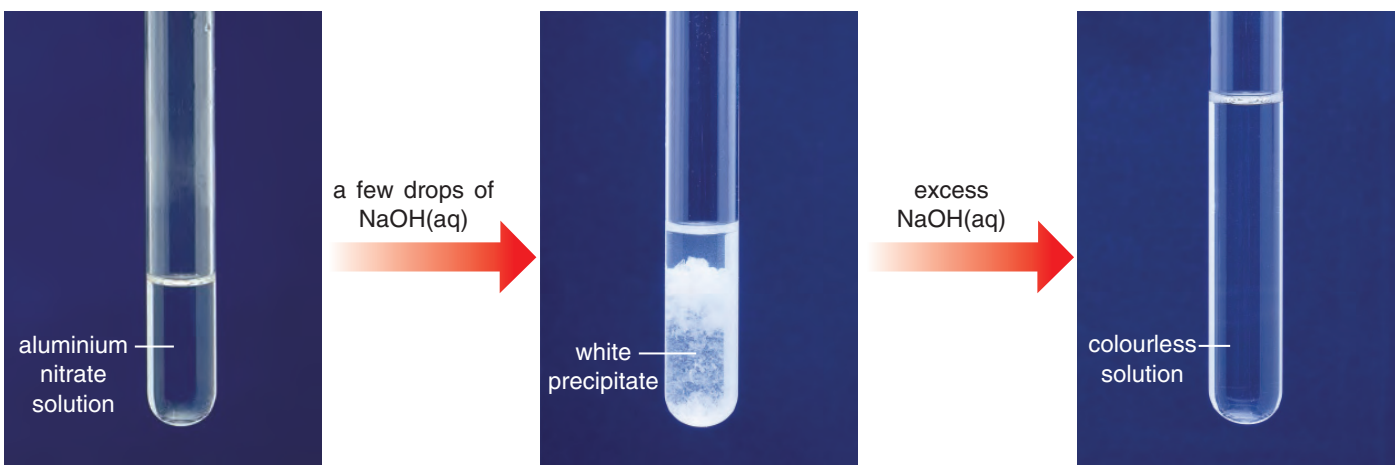


Fig. 53.12 The reaction of aluminium nitrate solution with dilute aqueous solution of sodium hydroxide

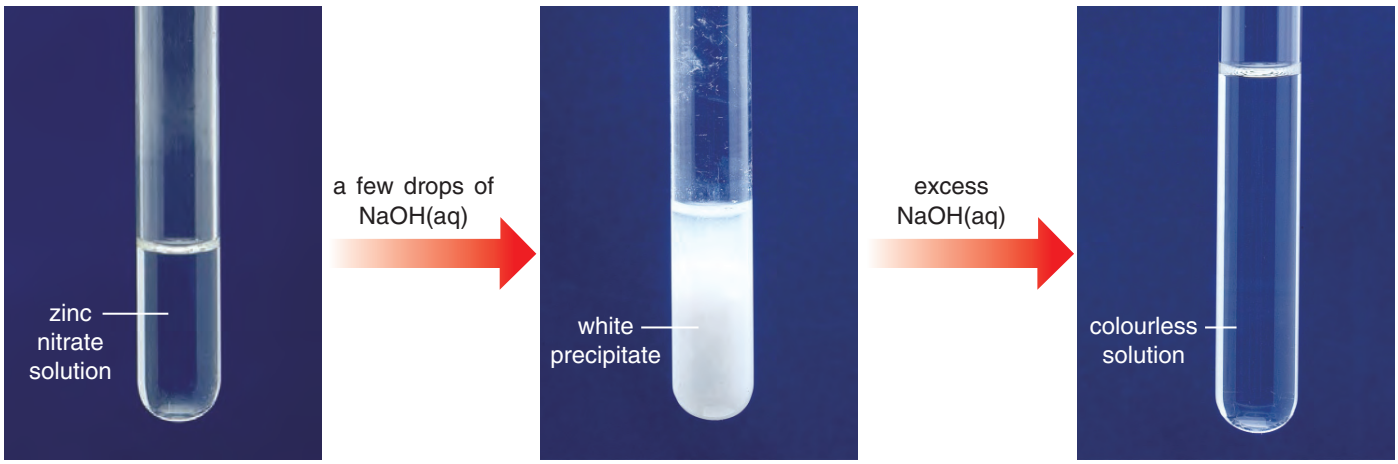


Fig. 53.13 The reaction of zinc nitrate solution with dilute aqueous solution of sodium hydroxide

53.10 Action of dilute aqueous ammonia N6

N6 Refer to page T23.

Like dilute aqueous solution of sodium hydroxide, dilute aqueous ammonia also precipitates insoluble hydroxides from solutions containing some cations. In some cases, the hydroxides dissolve in excess dilute aqueous ammonia due to the formation of complex ions*. Table 53.7 summarizes the observations when we add dilute aqueous ammonia to solutions containing some cations.

A solution containing potassium ions, sodium ions, ammonium ions or calcium ions does not give a precipitate with dilute aqueous ammonia*.

During the test, prepare a solution if the sample is soluble in water. Then add dilute aqueous ammonia to the sample solution. We can deduce the cation present in the sample based on the observations. N7

* We have discussed these reactions in Topic 4 Acids and Bases.

* Aqueous ammonia is a weak alkali. Its concentration of hydroxide ions is less than that in an aqueous solution of sodium hydroxide of the same concentration. The cations of some solutions form slightly soluble hydroxides (such as calcium hydroxide) with dilute aqueous ammonia. Hence these cations do not give a precipitate with dilute aqueous ammonia.

N7 Refer to page T24.

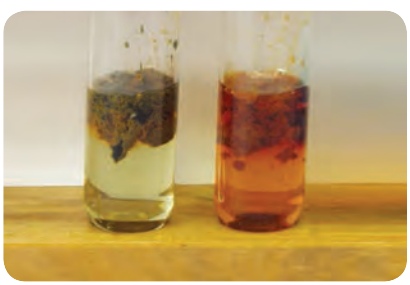


Fig. 53.14 Iron(II) hydroxide (left) and iron(III) hydroxide (right)

Table 53.7 Observations when dilute aqueous ammonia is added to solutions containing some cations

Adding $NH_3(aq)$ to solution containing	Colour of precipitate formed	Precipitate soluble in excess $NH_3(aq)$?	Ionic equations(s)
Magnesium ions, Mg^{2+}	white	✘	$Mg^{2+}(aq) + 2OH^-(aq) \longrightarrow Mg(OH)_2(s)$
Aluminium ions, Al^{3+}	white	✘	$Al^{3+}(aq) + 3OH^-(aq) \longrightarrow Al(OH)_3(s)$
Lead(II) ions, Pb^{2+}	white	✘	$Pb^{2+}(aq) + 2OH^-(aq) \longrightarrow Pb(OH)_2(s)$
Iron(II) ions, Fe^{2+}	green	✘	$Fe^{2+}(aq) + 2OH^-(aq) \longrightarrow Fe(OH)_2(s)$ CE 2006 Paper 2 Q21
Iron(III) ions, Fe^{3+}	reddish brown	✘	$Fe^{3+}(aq) + 3OH^-(aq) \longrightarrow Fe(OH)_3(s)$ CE 2002 Paper 2 Q37
Copper(II) ions, Cu^{2+}	pale blue	✔ (a deep blue solution forms)	$Cu^{2+}(aq) + 2OH^-(aq) \longrightarrow Cu(OH)_2(s)$ $Cu(OH)_2(s) + 4NH_3(aq) \longrightarrow [Cu(NH_3)_4]^{2+}(aq) + 2OH^-(aq)$ AL 1999 Paper 2 Q2d(ii) AL 2010 Paper 1 Q3(b)(ii)
Silver ions, Ag^+	brown	✔ (a colourless solution forms)	$2Ag^+(aq) + 2OH^-(aq) \longrightarrow Ag_2O(s) + H_2O(l)$ $Ag_2O(s) + H_2O(l) + 4NH_3(aq) \longrightarrow 2[Ag(NH_3)_2]^+(aq) + 2OH^-(aq)$
Zinc ions, Zn^{2+}	white	✔ (a colourless solution forms)	$Zn^{2+}(aq) + 2OH^-(aq) \longrightarrow Zn(OH)_2(s)$ AL 2005 Paper 2 Q1(c) $Zn(OH)_2(s) + 4NH_3(aq) \longrightarrow [Zn(NH_3)_4]^{2+}(aq) + 2OH^-(aq)$

N8 Refer to page T24.



Fig. 53.15 The reaction of copper(II) nitrate solution with dilute aqueous ammonia

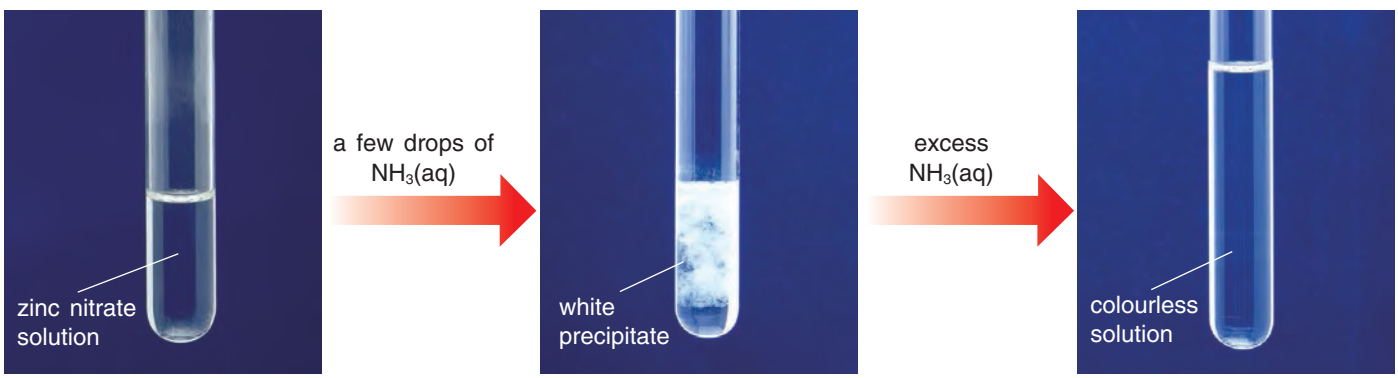


Fig. 53.16 The reaction of zinc nitrate solution with dilute aqueous ammonia



53.3 Identifying the cation present in a white salt

A white salt X contains only one cation. X is soluble in water and its solution gives a white precipitate with dilute aqueous solution of sodium hydroxide. The precipitate is insoluble in excess alkali.

a) Suggest TWO possible cations that may be present in X.

b) Give a test you may conduct to find out which of the two cations suggested in (a) is present in X.

Solution

a) Calcium ions or magnesium ions

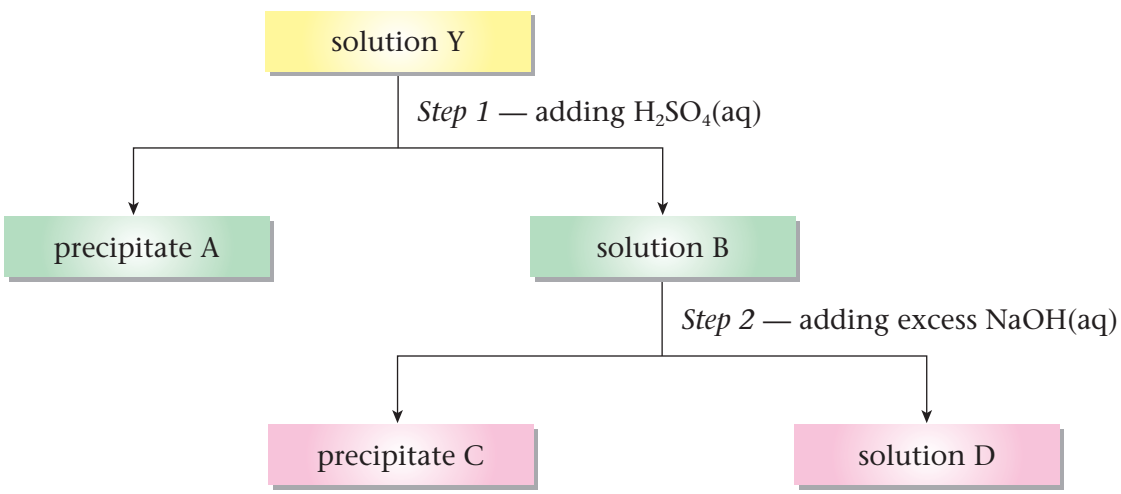
b) Conduct a flame test. A calcium salt will give a brick-red flame. Magnesium salt will not give a characteristic flame colour.

53.4 Identifying cations in a sample by studying a flow diagram

An aqueous solution Y contains the following cations:

- ammonium ions, $NH_4^+(aq)$
- iron(III) ions, $Fe^{3+}(aq)$
- lead(II) ions, $Pb^{2+}(aq)$

The following flow diagram outlines some tests that can be used to detect the presence of the cations in Y.



a) i) Name A.

ii) Write an ionic equation for the formation of A in *Step 1*.

iii) Suggest an experimental method that can be used to separate B from A.

b) i) Name C.

ii) Write an ionic equation for the formation of C in *Step 2*.

c) D still contains one of the above-mentioned cations. Suggest how you can show the presence of the cation in D. Explain your answer.

d) Based on the above information, suggest a colour for Y. Explain your answer.

Solution

a) i) Lead(II) sulphate

ii) $Pb^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow PbSO_4(s)$

iii) Filtration

b) i) Iron(III) hydroxide

ii) $Fe^{3+}(aq) + 3OH^-(aq) \longrightarrow Fe(OH)_3(s)$

c) The presence of ammonium ions can be shown by warming D. An alkaline gas will be evolved.

d) Y is yellow-brown in colour due to the presence of iron(III) ions. All the other ions are colourless.



Checkpoint

Refer to page T25.

1 Suggest a chemical test to distinguish each of the following pairs of chemicals. State the observable change(s) and write appropriate equation(s) for the reaction(s) involved.

a) Aqueous solutions of magnesium sulphate and zinc sulphate

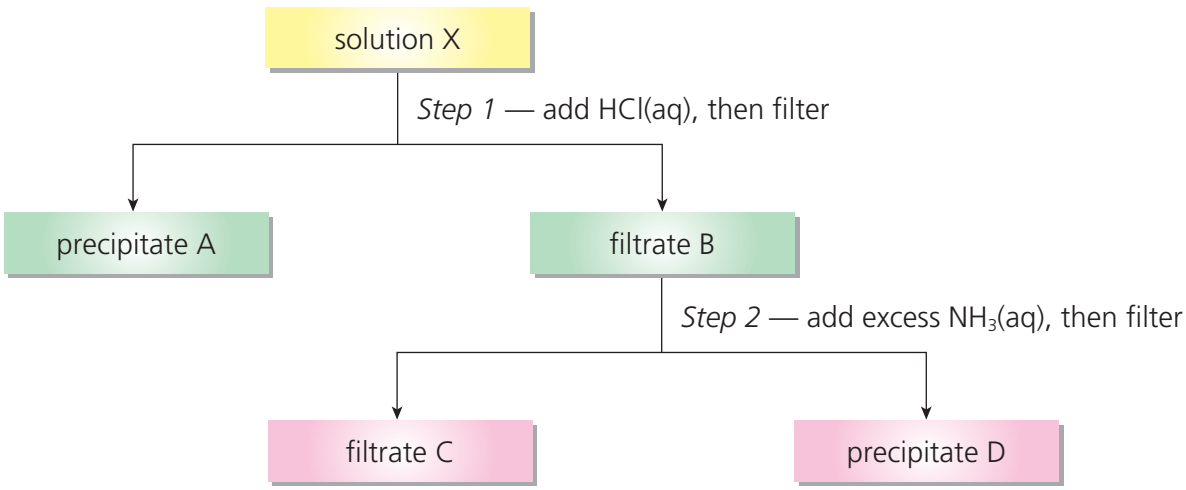
b) Solid ammonium chloride and solid potassium chloride

c) Aqueous solutions of copper(II) nitrate and lead(II) nitrate

2 An aqueous solution X contains the following cations:

- iron(II) ions, $Fe^{2+}(aq)$
- silver ions, $Ag^{+}(aq)$
- zinc ions, $Zn^{2+}(aq)$

The following flow diagram outlines some tests that can be used to detect the presence of the cations in X.



a) i) Name A.

ii) Write an ionic equation for the formation of A in *Step 1*.

b) What cation is present in C? Explain your answer.

c) Based on the above information, suggest a colour for X. Explain your answer.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Problem Solving

Refer to page T26.

There are five unlabelled reagent bottles, each containing one of the following white solids:

- ammonium chloride
- lead(II) chloride
- magnesium chloride
- sodium chloride
- sugar

Suggest how you would carry out tests to distinguish the five solids from one another.

53.11 Identifying the cation in an unknown sample in the laboratory

After studying the action of various reagents on cations, you are going to carry out chemical tests in the laboratory and identify the cation in a sample based on your observations.

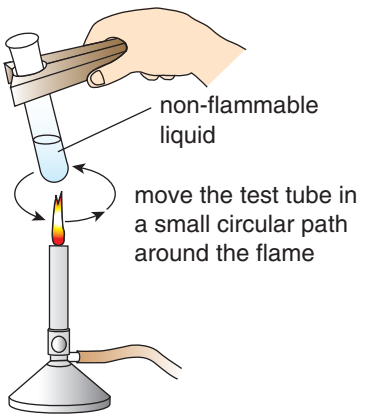


Fig. 53.17 Heating a non-flammable liquid in a test tube*

Conducting chemical tests

Observe the following rules when conducting chemical tests in the laboratory.

- Always follow the instructions carefully. For example, follow instructions for the extent of heating, e.g. warm gently or heat strongly.
- Use small amounts of solids to give dilute solutions. If you add a lot of a solid, you can easily hide the effect you are looking for.
- When heating, always start by warming gently. Move the test tube around the flame to keep the contents of the test tube moving (Fig. 53.17). Make sure that the contents are heated evenly.
- Be prepared to test for any gas evolved during tests; always get ready all reagents, indicator papers, wooden splint, delivery tube, etc.*.

Making and recording observations

Write down all that you observe, including

- colour of solid / precipitate / solution;
- colour changes* upon heating / cooling / when left standing in the air;
- temperature changes;
- solid dissolving;
- precipitate forming;
- colour and odour of any gas or vapour evolved;
- any effervescence occurring; and
- results of tests for identifying gases or vapours.

* When heating a non-flammable liquid in a test tube, heat the top part of the liquid first.

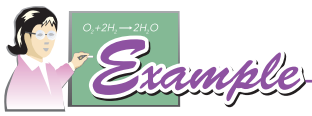
* We will discuss specific tests for common gases and vapours later in this unit.

* If there is a colour change, always record the colour before and after the test.

Take care not to confuse observations and deductions. Suppose dilute aqueous ammonia is added until excess to a blue sample solution (aqueous solution of copper(II) sulphate). The observation is that a blue precipitate forms initially, but the precipitate dissolves in excess aqueous ammonia to give a deep blue solution. The deduction that the blue precipitate is copper(II) hydroxide is not an observation.

Making deductions

Interpret the observations and make deductions about ions that may be present in the sample under test.



53.5 Identifying the cation in a sample from a record of observations of tests

Consider an aqueous solution X containing one cation. Some tests are carried out on X and the observations shown in the table below.

a) Complete the last column of deductions that can be made based on the observation of each test.

b) Identify, with explanation, the cation in X.

Solution

a)

	Test	Observation	Deduction
1	a) To a portion of X, add dilute aqueous solution of sodium hydroxide until a change is observed.	A white precipitate is formed.	Aluminium ions, lead(II) ions or zinc ions may be present.
	b) Add excess aqueous solution of sodium hydroxide to the mixture from (a).	The precipitate dissolves to give a colourless solution.	
2	a) To a portion of X, add dilute aqueous ammonia until a change is observed.	A white precipitate is formed.	Aluminium ions, lead(II) ions or magnesium ions may be present.
	b) Add excess dilute aqueous ammonia to the mixture from (a).	The precipitate remains insoluble.	
3	To a portion of X, add dilute hydrochloric acid.	A white precipitate is formed.	Lead(II) ions or silver ions may be present.

b) From Test 1, X gives a white precipitate with dilute aqueous solution of sodium hydroxide and the precipitate is soluble in excess alkali. A solution containing aluminium ions, lead(II) ions or zinc ions would give such a result.

From Test 2, X gives a white precipitate with dilute aqueous ammonia and the precipitate does not dissolve in excess alkali. As a solution containing zinc ions would give a white precipitate that is soluble in excess alkali, thus we can eliminate zinc ions from the list. This leaves aluminium ions and lead(II) ions on the list.

The observation in Test 3 shows that X contains lead(II) ions as it gives a white precipitate (lead(II) chloride) with dilute hydrochloric acid.

Therefore we can conclude that X contains lead(II) ions.

In the following activity, you are going to identify the cation in an unknown solid sample based on the chemical tests discussed above.

Video
Flame test



Activity 53.1

Identifying the cation in an unknown solid sample.

53.12 Devising a scheme to separate the cations in an aqueous solution

Suppose we are given an aqueous solution containing five cations:

$Ag^+(aq)$, $Ca^{2+}(aq)$, $Fe^{2+}(aq)$, $Na^+(aq)$, $Pb^{2+}(aq)$

How can we separate the cations in the solution? We need to look at the properties of the cations:

- $Ag^+(aq)$ ions and $Pb^{2+}(aq)$ ions form insoluble chlorides;
- $Ca^{2+}(aq)$ ions form an insoluble sulphate; and
- $Fe^{2+}(aq)$ ions form an insoluble hydroxide.

Study the flow diagram shown in Fig. 53.18 to see how we can separate the cations based on their properties.

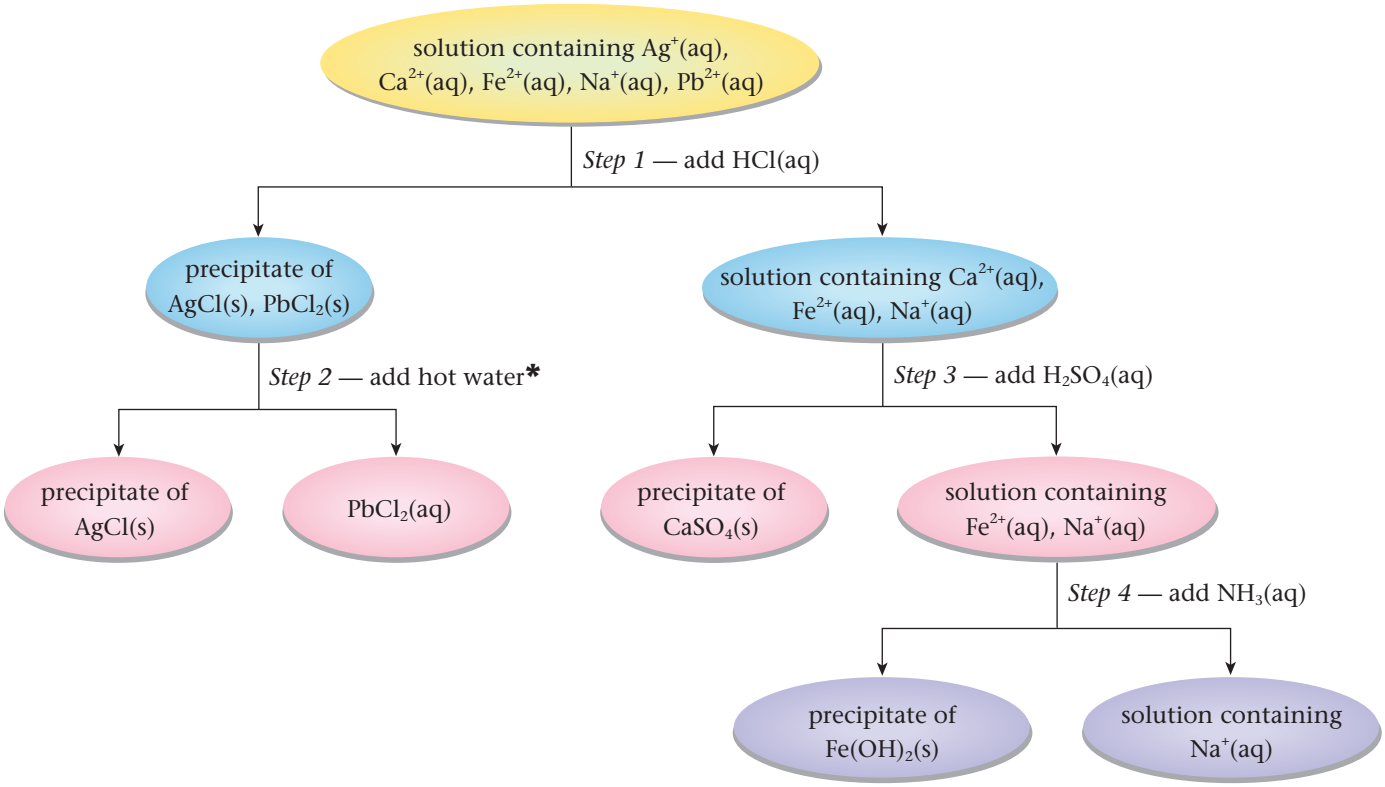
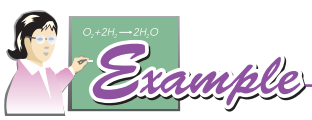


Fig. 53.18 A separation scheme for an aqueous solution containing five cations: $Ag^+(aq)$, $Ca^{2+}(aq)$, $Fe^{2+}(aq)$, $Na^+(aq)$, $Pb^{2+}(aq)$

* $PbCl_2(s)$ is insoluble in cold water, but soluble in hot water. Thus, we can separate $PbCl_2(s)$ from AgCl(s) by using hot water.

This is just one separation scheme for the cations in a solution. We may separate the cations using other separation schemes, depending on the cations present in the given sample and the reagents available.



53.6 Devising a separation scheme for an aqueous solution containing four cations

Suppose you are given an aqueous solution containing four cations:

$Al^{3+}(aq)$, $Cu^{2+}(aq)$, $Pb^{2+}(aq)$, $Zn^{2+}(aq)$

You are provided with the following reagents:

6 mol dm^{-3} HCl(aq) 6 mol dm^{-3} $NH_3(aq)$ 6 mol dm^{-3} NaOH(aq)

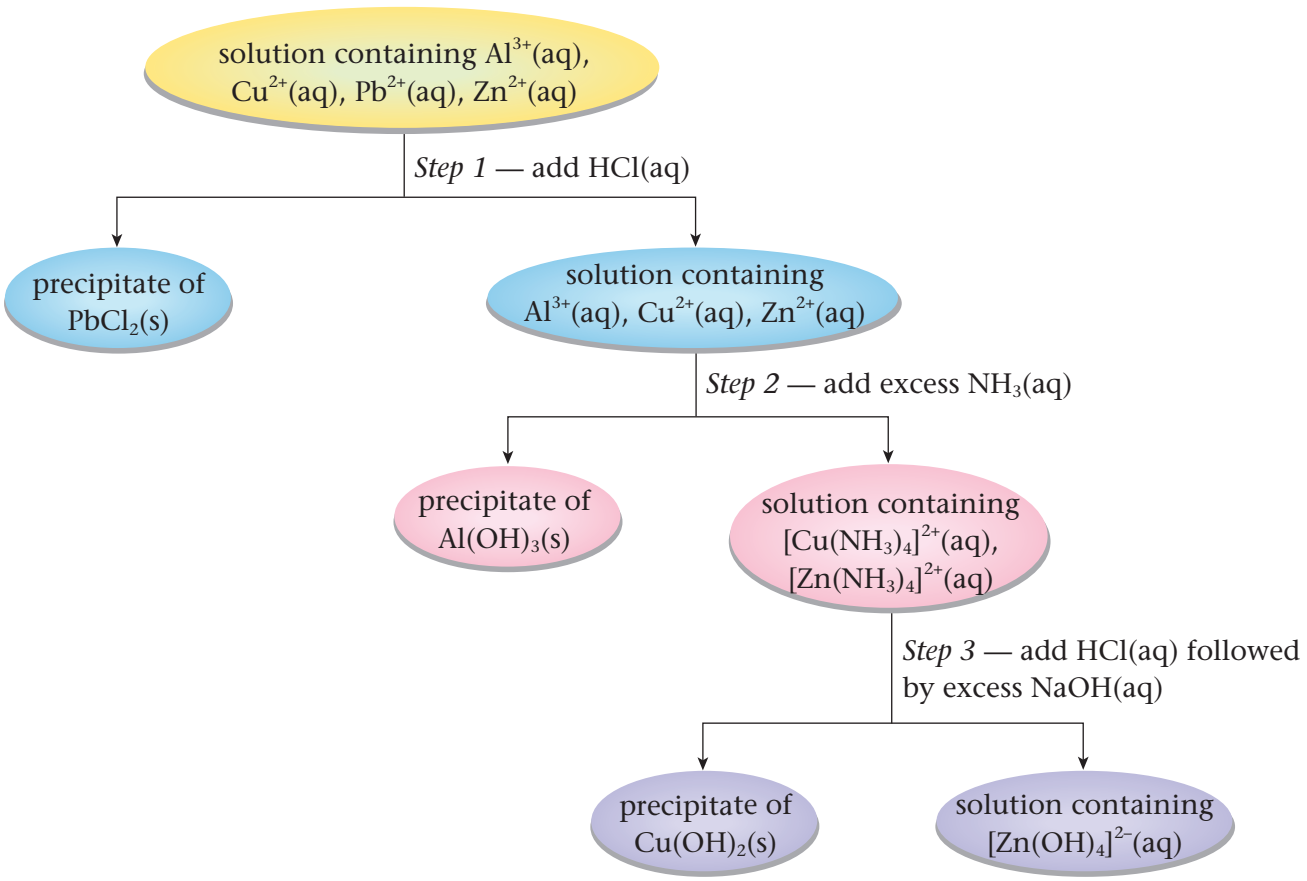
Devise a separation scheme for the cations in the solution. Explain the rationale of your scheme and present it in the form of a flow diagram.

Solution

We need to look at the properties of the cations:

- $Pb^{2+}(aq)$ ions form an insoluble chloride;
- $Al^{3+}(aq)$ ions form an insoluble hydroxide with dilute aqueous ammonia;
- $Cu^{2+}(aq)$ ions and $Zn^{2+}(aq)$ ions form insoluble hydroxides with dilute aqueous ammonia and the hydroxides are soluble in excess aqueous ammonia;
- $Cu^{2+}(aq)$ ions form an insoluble hydroxide with dilute aqueous solution of sodium hydroxide;
- $Zn^{2+}(aq)$ ions form an insoluble hydroxide with dilute aqueous solution of sodium hydroxide and the hydroxide is soluble in excess alkali.

The following flow diagram summarizes the separation scheme for the cations.



Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Problem Solving

Refer to page T27.

Suppose you are given an aqueous solution containing four cations:

$Ag^+(aq)$, $Ca^{2+}(aq)$, $K^+(aq)$, $Mg^{2+}(aq)$

Devise a separation scheme for the cations in the solution. Explain the rationale of your scheme and present it in the form of a flow diagram.

53.13 Specific tests for identifying common gases and vapours

N9 Refer to page T28.

When identifying the cation and / or anion in a sample, we need to conduct various tests and identify any gas or vapour given off. N9 Therefore it is important to learn specific tests for identifying common gases and vapours.

We can identify gases and vapours given off during any test using the following methods:

- from their colours;
- from their odours*;
- action on moist litmus paper; and
- chemical test.

* **Caution:** Smell any gas or vapour by fanning the gas or vapour gently towards the nose, but NOT by direct inhaling.

Ammonia

* **Caution:** Ammonia is toxic. Perform experiments involving ammonia inside a fume cupboard.

Ammonia* is a colourless, alkaline gas with a pungent smell. It turns moist red litmus paper blue.

Ammonia reacts with hydrogen chloride to give ammonium chloride according to the following equation:

$$NH_3(g) + HCl(g) \longrightarrow NH_4Cl(s)$$

Dense white fumes (ammonium chloride) can be observed. This serves as a test for ammonia. We can also test hydrogen chloride with ammonia.



Fig. 53.19 Ammonia turns moist red litmus paper blue



Fig. 53.20 Testing $NH_3(g)$ with HCl(g)

Video
Action of solid calcium hydroxide on ammonium chloride

Carbon dioxide

Carbon dioxide is a colourless, weakly acidic gas. It turns limewater* milky due to the formation of insoluble white calcium carbonate.

$$CO_2(g) + Ca(OH)_2(aq) \longrightarrow CaCO_3(s) + H_2O(l)$$

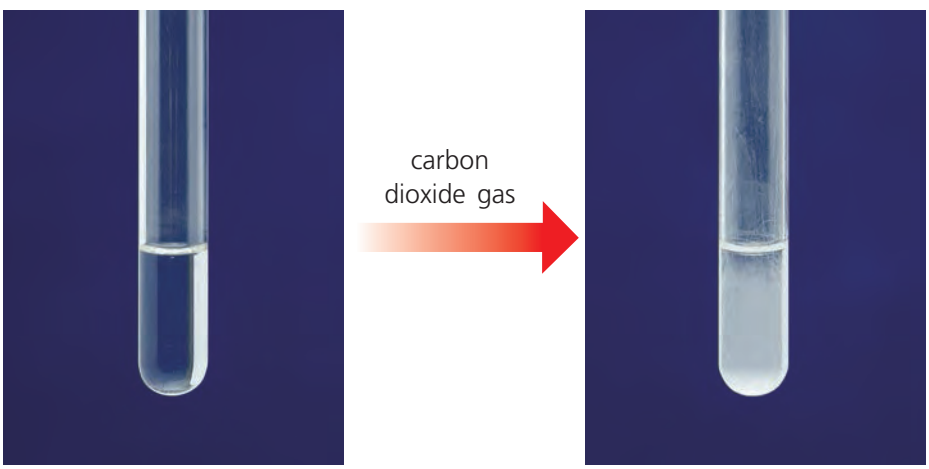


Fig. 53.21 Carbon dioxide turns limewater milky

* **Caution:** Sulphur dioxide also turns limewater milky due to the formation of insoluble white calcium sulphite.
$SO_2(g) + Ca(OH)_2(aq) \longrightarrow CaSO_3(s) + H_2O(l)$

Chlorine

Chlorine* is a greenish yellow acidic gas with a pungent smell. It dissolves in water to form a pale greenish yellow solution. Chlorine reacts with water, forming hydrochloric acid and hypochlorous acid. The solution formed is very acidic.

$$\underset{}{Cl_2(g)} + \underset{}{H_2O(l)} \rightleftharpoons \underset{\text{hydrochloric acid}}{HCl(aq)} + \underset{\text{hypochlorous acid}}{HOCl(aq)}$$

* **Caution:** Chlorine is toxic. Perform experiments involving chlorine inside a fume cupboard.

Aqueous chlorine can have a bleaching action due to the oxidizing action of the hypochlorite ions in the hypochlorous acid. The ions give up the oxygen to the coloured dye, making the dye colourless.

$$\underset{\text{colourless}}{OCl^-(aq)} + \underset{\text{coloured}}{\text{dye}} \longrightarrow \underset{\text{colourless}}{Cl^-(aq)} + \overbrace{\underset{\text{colourless}}{(O + \text{dye})}}^{\text{oxidized form}}$$

Chlorine turns moist blue litmus paper red and then bleaches it.

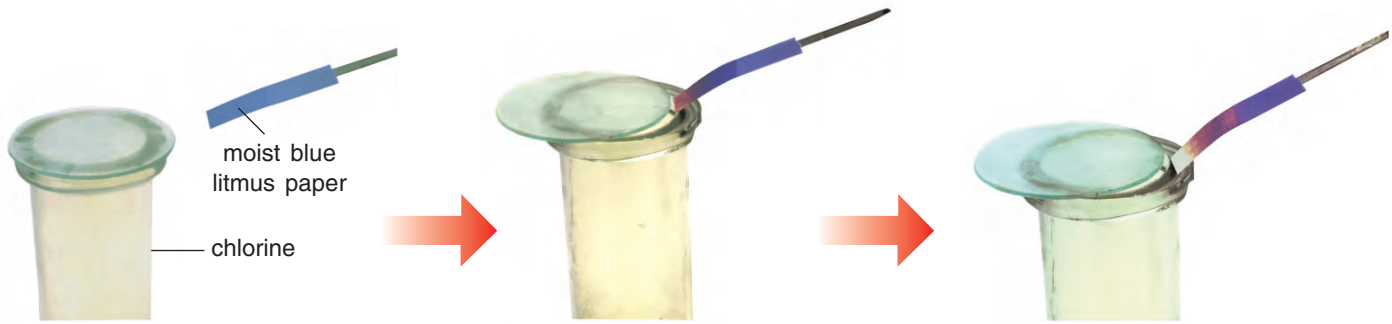


Fig. 53.22 Chlorine turns moist blue litmus paper red and then bleaches it

Sulphur dioxide

* **Caution:** Sulphur dioxide is toxic. Perform experiments involving sulphur dioxide inside a fume cupboard.

Sulphur dioxide* is a colourless gas with a smell characteristic of burning sulphur.

Sulphur dioxide can turn a piece of filter paper soaked with acidified potassium dichromate solution from orange to green.

AL 2005 Paper 1 Q7(c)(iii)

$$\underset{\text{orange}}{Cr_2O_7^{2-}(aq)} + 3SO_2(aq) + 2H^+(aq) \longrightarrow \underset{\text{green}}{2Cr^{3+}(aq)} + 3SO_4^{2-}(aq) + H_2O(l)$$

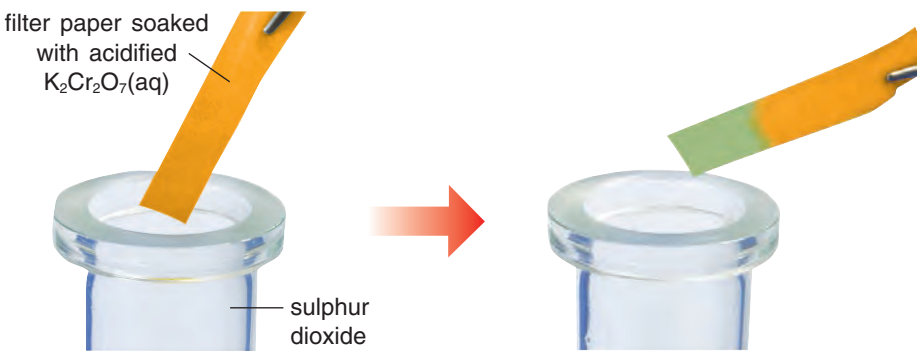


Fig. 53.23 Sulphur dioxide turns filter paper soaked with acidified potassium dichromate solution from orange to green

Table 53.8 summarizes specific tests for some common gases and vapours.

Table 53.8 Specific tests for some common gases and vapours

Gas or vapour	Colour	Test	Observation
Ammonia	Colourless	Insert a piece of moist red litmus paper into the gas.	It turns moist red litmus paper blue.
		Dip a glass rod in dilute hydrochloric acid and insert it into the gas.	Dense white fumes are formed.
Carbon dioxide	Colourless	Bubble the gas through limewater.	The limewater turns milky.
Chlorine	Greenish yellow	Insert a piece of moist blue litmus paper into the gas.	Chlorine turns the moist blue litmus paper red and then bleaches it.
Bromine	Reddish brown	Insert a piece of moist blue litmus paper into the vapour.	Bromine turns the moist blue litmus paper red and then bleaches it.
Iodine	Purple	Insert a piece of moist starch paper into the vapour.	The moist starch paper turns blue-black*.
Hydrogen	Colourless	Insert a burning splint into the gas.	A 'pop' sound is heard.
Oxygen	Colourless	Insert a glowing splint into the gas.	The glowing splint relights.
Hydrogen halides	Colourless	Dip a glass rod in aqueous ammonia and insert it into the gas.	Dense white fumes are formed*.
Sulphur dioxide	Colourless	Insert a piece of filter paper soaked with acidified potassium dichromate solution into the gas.	The paper turns from orange to green.
Water vapour	Colourless	Insert a piece of blue cobalt(II) chloride paper into the vapour*.	The blue cobalt(II) chloride turns pink.

DSE 2014 Paper 2 Q3(a)(i)(I)

AL 1999 Paper 1 Q7(d)
AL 2000 Paper 1 Q3(c)(iii)

Video
Action of water, ethanol and oil on dry cobalt(II) chloride paper

* Iodine forms a dark blue complex with starch.

* Hydrogen bromide may be accompanied by reddish brown bromine vapour. Hydrogen iodide may be accompanied by purple iodine vapour.

* The evolution of water vapour may be detected by looking out for water droplets at the cooler end of the test tube.

53.14 Identifying the anion in a sample

We shall use a number of anions to illustrate the general approach for identifying the anion in a sample:

- chloride ions, Cl^-;
- bromide ions, Br^-;
- iodide ions, I^-;
- carbonate ions, CO_3^{2-};
- hypochlorite ions, OCl^-;
- sulphite ions, SO_3^{2-};
- sulphate ions, SO_4^{2-}.

We will look at each of the following tests in turn:

- action of heat;
- action of dilute hydrochloric acid;
- action of concentrated sulphuric acid;
- action of aqueous solution of silver nitrate.

53.15 Action of heat

Most solids undergo changes when heated. The identity of the gas given off, if any, provides a clue to the anion in the sample.

Carbonates

All carbonates decompose on heating (except sodium carbonate and potassium carbonate). Carbon dioxide is given off and the corresponding oxide remains, except in the case of ammonium carbonate where ammonia and water vapour are also given off.

e.g. $CuCO_3(s) \xrightarrow{\text{heat}} CuO(s) + CO_2(g)$

$(NH_4)_2CO_3(s) \xrightarrow{\text{heat}} 2NH_3(g) + CO_2(g) + H_2O(g)$

Sulphates

Sulphates are generally more stable to heat than carbonates. However, certain sulphates may decompose on strong heating.

For example, iron(II) sulphate decomposes to give iron(III) oxide, sulphur trioxide and sulphur dioxide.

$$2FeSO_4(s) \xrightarrow{\text{heat}} Fe_2O_3(s) + SO_3(s) + SO_2(g)$$ N10

N10 Refer to page T30.

53.16 Action of dilute hydrochloric acid

During the test, add the dilute hydrochloric acid to a little of the solid sample or its aqueous solution. Then test for any gas given off*. N11 N12 N13

Gas given off

A sample containing carbonate ions (CO_3^{2-}) gives carbon dioxide with dilute hydrochloric acid. N14

$$CO_3^{2-}(aq) + 2H^+(aq) \longrightarrow CO_2(g) + H_2O(l)$$

CE 2000 Paper 2 Q22
CE 2005 Paper 2 Q26
ASL 2008 Paper 2 Q1(c)
AL 2008 Paper 2 Q2(c)

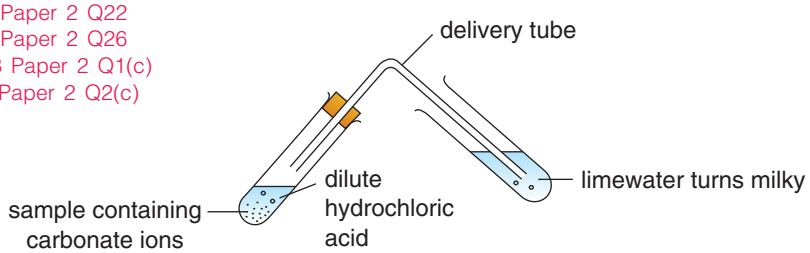


Fig. 53.24 A sample containing carbonate ions gives carbon dioxide with dilute hydrochloric acid

A solution containing hypochlorite ions (OCl^-) gives chlorine with dilute hydrochloric acid.

$$OCl^-(aq) + 2H^+(aq) + Cl^-(aq) \longrightarrow Cl_2(g) + H_2O(l)$$

CE 2005 Paper 1 Q12

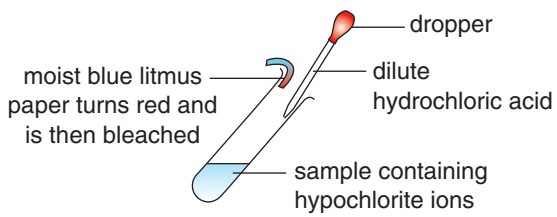


Fig. 53.25 A sample containing hypochlorite ions gives chlorine with dilute hydrochloric acid

* Functions of dilute sulphuric acid are similar to those of dilute hydrochloric acid. A sample containing CO_3^{2-} ions, OCl^- ions or SO_3^{2-} ions gives the same gas if dilute sulphuric acid is used instead of dilute hydrochloric acid in the test.

Video
Test for carbonate ions

N11 Refer to page T31.

N12 Adding $H_2SO_4(aq)$ or $HCl(aq)$ to $K_2CrO_4(aq)$

Examination questions often ask about the colour change when $H_2SO_4(aq)$ or $HCl(aq)$ is added to $K_2CrO_4(aq)$.

Yellow $K_2CrO_4(aq)$ turns orange upon the addition of $H_2SO_4(aq)$ or $HCl(aq)$.

$$\underset{\text{yellow}}{2CrO_4^{2-}(aq)} + 2H^+(aq) \rightleftharpoons \underset{\text{orange}}{Cr_2O_7^{2-}(aq)} + H_2O(l)$$

A green solution results when $FeSO_4(aq)$ is added to the orange solution.

$$6Fe^{2+}(aq) + \underset{\text{orange}}{Cr_2O_7^{2-}(aq)} + 14H^+(aq) \longrightarrow 6Fe^{3+}(aq) + 2Cr^{3+}(aq) + 7H_2O(l)$$

green

N13 The reactions of concentrated hydrochloric acid are similar to those of dilute hydrochloric acid but are more vigorous.

N14 Refer to page T31.

N15 A solution containing thiosulphate ions ($S_2O_3^{2-}$) gives sulphur dioxide and sulphur deposits with dilute hydrochloric acid.
$S_2O_3^{2-}(aq) + 2H^+(aq) \longrightarrow SO_2(g) + H_2O(l) + S(s)$

A solution containing sulphite ions (SO_3^{2-}) gives sulphur dioxide with dilute hydrochloric acid. **N15**

$$SO_3^{2-}(aq) + 2H^+(aq) \longrightarrow SO_2(g) + H_2O(l)$$

CE 2005 Paper 2 Q26
CE 2006 Paper 2 Q15
CE 2008 Paper 2 Q3

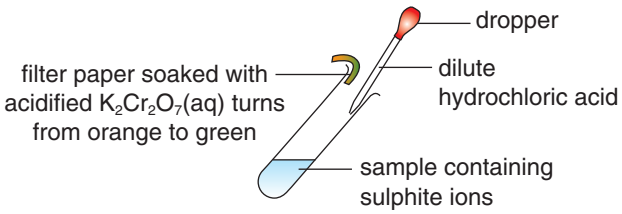


Fig. 53.26 A sample containing sulphite ions gives sulphur dioxide with dilute hydrochloric acid

N16 Refer to page T32.

Video
Test for sulphate ions

Test for sulphate ions N16

We can use aqueous solution of barium chloride or barium nitrate to detect anions that form insoluble barium salts.

N17 Examination questions often ask about identifying a solution of sulphate ions by using an aqueous solution of barium chloride.

Sulphate ions, sulphite ions and carbonate ions all form a white precipitate with an aqueous solution of barium chloride. **N17**

$$Ba^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow BaSO_4(s)$$

$$Ba^{2+}(aq) + SO_3^{2-}(aq) \longrightarrow BaSO_3(s)$$

$$Ba^{2+}(aq) + CO_3^{2-}(aq) \longrightarrow BaCO_3(s)$$

CE 2004 Paper 1 Q2(c)
CE 2005 Paper 1 Q12
AL 2003 Paper 1 Q7(b)
AL 2005 Paper 2 Q1(c)

Barium sulphite is soluble in dilute hydrochloric acid with the evolution of sulphur dioxide.

$$BaSO_3(s) + 2H^+(aq) \longrightarrow Ba^{2+}(aq) + SO_2(g) + H_2O(l)$$

Barium carbonate is also soluble in dilute hydrochloric acid with the evolution of carbon dioxide.

$$BaCO_3(s) + 2H^+(aq) \longrightarrow Ba^{2+}(aq) + CO_2(g) + H_2O(l)$$

However, barium sulphate is insoluble in dilute hydrochloric acid.

Therefore to test for sulphate ions in a sample solution, we first add a few drops of aqueous solution of barium chloride, followed by dilute hydrochloric acid. Formation of a white precipitate that is insoluble in the acid shows the presence of sulphate ions.

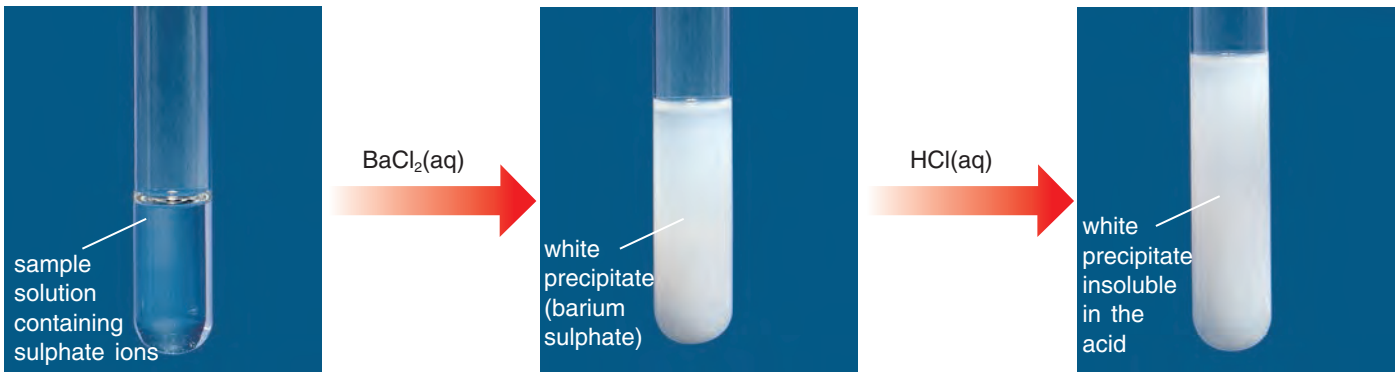


Fig. 53.27 Barium sulphate is insoluble in dilute hydrochloric acid



53.7 Studying some reactions starting from sulphur dioxide

The flow diagram below shows a series of reactions starting from sulphur dioxide gas.



a) i) Name X.

ii) Write an equation for the reaction between sulphur dioxide and an aqueous solution of barium chloride.

b) i) Name Y.

ii) Write an equation for the reaction between X and dilute hydrochloric acid to form Y.

c) Another colourless gas G gives the same sequence of observable changes as shown in the flow diagram. Name G.

Solution

a) i) Barium sulphite

ii) $Ba^{2+}(aq) + SO_2(g) + H_2O(l) \longrightarrow BaSO_3(s) + 2H^+(aq)$

b) i) Barium chloride

ii) $BaSO_3(s) + 2HCl(aq) \longrightarrow BaCl_2(aq) + SO_2(g) + H_2O(l)$

c) G is carbon dioxide.

53.8 Identifying three solids

Three reagent bottles labelled A, B and C each contains one of the following solids: calcium hydroxide, magnesium carbonate and sodium sulphate.

Two tests were carried out on the solid from each reagent bottle and the results are shown below:

Test	Solid from reagent bottle		
	A	B	C
Action of heat	no observable change	no visible change, but the residue after heating weighed less than the original solid	?
Action of dilute hydrochloric acid	no observable change	?	effervescence occurred

a) Name the solid in bottle A.

b) i) Why was there a loss in weight when the solid in bottle B was heated?

ii) Write an equation for the reaction that occurred when the solid in bottle B was heated.

iii) Name the type of reaction that occurred when the solid in bottle B reacted with dilute hydrochloric acid.

c) i) Name the solid in bottle C. Explain your answer.

ii) Write an equation for the reaction that occurred when the solid in bottle C was heated.

Solution

a) Sodium sulphate

b) i) The solid in bottle B decomposed on heating and a vapour escaped. So there was a loss in weight.

ii) $Ca(OH)_2(s) \longrightarrow CaO(s) + H_2O(g)$

iii) Neutralization

c) i) The solid in bottle C was magnesium carbonate as effervescence occurred (carbon dioxide gas was evolved) when it reacted with dilute hydrochloric acid.

ii) $MgCO_3(s) \longrightarrow MgO(s) + CO_2(g)$

53.17 Action of concentrated sulphuric acid

* We have discussed the oxidizing property of concentrated sulphuric acid in Topic 5 Redox Reactions, Chemical cells and Electrolysis.

Concentrated sulphuric acid possesses an oxidizing property*. This property is often observed in the reaction of the acid with the sample under test.

During the test, add a few drops of concentrated sulphuric acid to wet the solid sample in a dry test tube. Warm if no reaction occurs in the cold. Then test for any gas given off and notice any colour change.

AL 2011 Paper 1 Q7(b)(ii)

Table 53.9 summarizes gases that may be given off when concentrated sulphuric acid is added to a sample containing chloride ions (Cl^-), bromide ions (Br^-) or iodide ions (I^-).

N18 Refer to page T32.

CE 2004 Paper 1 Q2(a)

ASL 1999 Q12(b)(i),(ii)

Table 53.9 Gases given off upon the action of concentrated sulphuric acid on a sample containing chloride ions (Cl^-), bromide ions (Br^-) or iodide ions (I^-) N18

Observation about gas(es) given off upon the action of concentrated H_2SO_4	Gas(es) given off	Anions in sample
Steamy fumes that form dense white fumes with aqueous ammonia	hydrogen chloride	chloride ions, Cl^-
Reddish brown vapour that forms dense white fumes with aqueous ammonia; condenses to a reddish brown liquid	hydrogen bromide + bromine vapour + sulphur dioxide	bromide ions, Br^-
Purple vapour condenses to a black solid	iodine vapour + sulphur dioxide + hydrogen sulphide	iodide ions, I^-

CE 2005 Paper 1 Q12
CE 2006 Paper 1 Q4
CE 2007 Paper 2 Q1
CE 2009 Paper 2 Q23
CE 2011 Paper 2 Q41
ASL 2001 Q7
AL 2003 Paper 1 Q7(b)
AL 2004 Paper 1 Q7(a)(ii)
AL 2005 Paper 2 Q1(c)

ASL 1999 Q12(b)(i),(ii)
AL 2000 Paper 1 Q3(c)(i)

53.18 Action of aqueous solution of silver nitrate

DSE 2015 Paper 2 Q3(a)(i)
AL 2011 Paper 1 Q7(b)(ii)

We can also identify chloride, bromide or iodide ions in a sample by the action of aqueous solution of silver nitrate. N19

During the test, dissolve the sample in water. Add an equal amount of dilute nitric acid*. Then add several drops of aqueous solution of silver nitrate. Formation of a precipitate indicates the presence of halide ions. N20

$$Ag^+(aq) + Cl^-(aq) \longrightarrow \underset{\text{white precipitate}}{AgCl(s)}$$

AL 2011 Paper 1 Q7(b)(ii)

* Acidification with nitric acid can eliminate carbonate ions and sulphite ions, both of which would interfere with the test by giving precipitates with $Ag^+(aq)$ ions.

$$2Ag^+(aq) + CO_3^{2-}(aq) \longrightarrow Ag_2CO_3(s)$$

$$2Ag^+(aq) + SO_3^{2-}(aq) \longrightarrow Ag_2SO_3(s)$$

Video
Test for chloride ions

N19 Aqueous solution of silver nitrate can also precipitate sulphate ions from concentrated sulphate solutions.

N20 Examination questions often ask about identifying a solution of chloride ions by using an aqueous solution of silver nitrate.

N21 Refer to page T33.

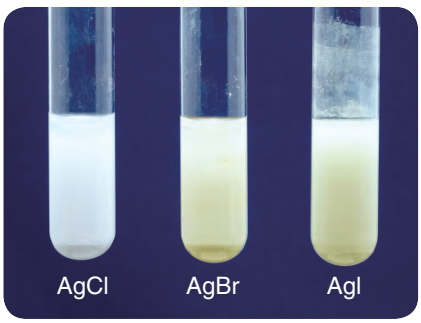


Fig. 53.28 The precipitates formed by adding dilute nitric acid and an aqueous solution of silver nitrate to solutions containing halide ions

$$Ag^+(aq) + Br^-(aq) \longrightarrow \underset{\text{creamy precipitate}}{AgBr(s)}$$

AL 2011 Paper 1 Q7(b)(ii)

$$Ag^+(aq) + I^-(aq) \longrightarrow \underset{\text{yellow precipitate}}{AgI(s)}$$

The silver chloride precipitate is readily soluble in aqueous ammonia to form a complex salt.

$$AgCl(s) + 2NH_3(aq) \longrightarrow [Ag(NH_3)_2]^+(aq) + Cl^-(aq)$$

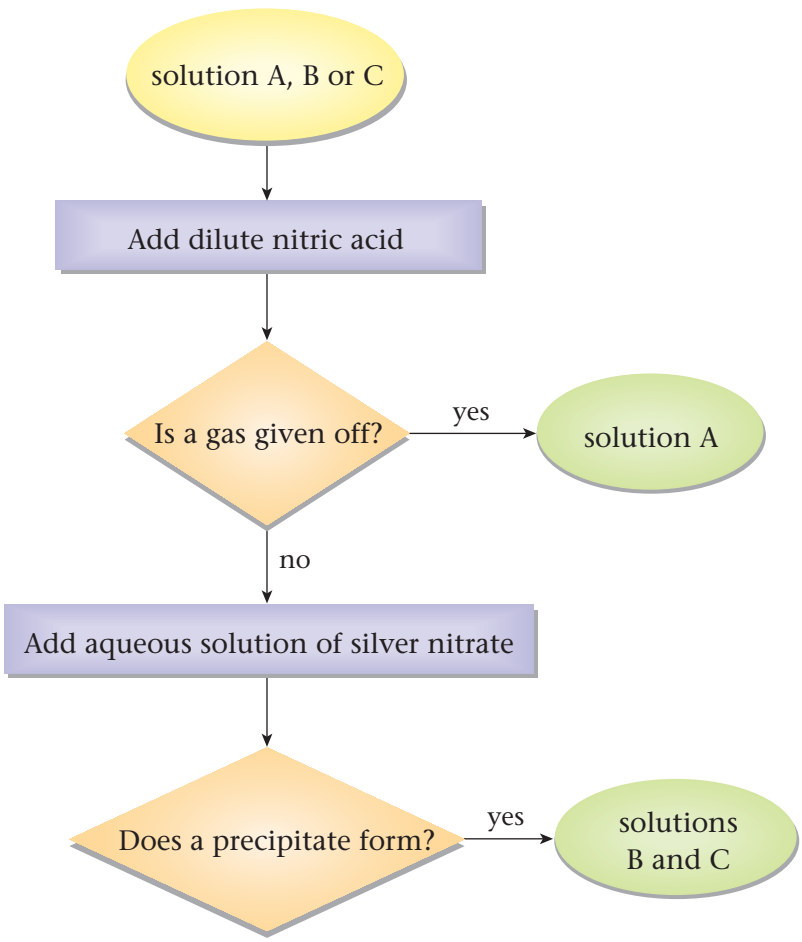
The silver bromide precipitate is only soluble in very concentrated aqueous ammonia while the silver iodide precipitate is insoluble in aqueous ammonia. N21



53.9 Identifying anions by studying a flow diagram

The flow diagram below outlines some tests that can be used to identify the anions in aqueous solutions A, B and C. Each of the solutions contains one of the following anions:

bromide ions, carbonate ions, iodide ions



a) Which anion does A contain? Write an equation for the reaction between A and dilute nitric acid.

b) The anions present in B and C cannot be distinguished easily using the above procedure. Suggest another chemical test to distinguish between them.

Solution

a) A contains carbonate ions.

$CO_3^{2-}(aq) + 2H^+(aq) \longrightarrow CO_2(g) + H_2O(l)$

CE 2009 Paper 1 Q7(b) AL 2011 Paper 1 Q7(b)(ii)

CE 2007 Paper 2 Q19
ASL 2000 Q2

b) Add aqueous chlorine to B and C separately.

- The solution containing bromide ions becomes yellow-brown in colour due to the formation of bromine.

  $Cl_2(aq) + 2Br^-(aq) \longrightarrow 2Cl^-(aq) + Br_2(aq)$

- The solution containing iodide ions becomes brown in colour due to the formation of iodine.

  $Cl_2(aq) + 2I^-(aq) \longrightarrow 2Cl^-(aq) + I_2(aq)$

N22

To show clearly that bromine and iodine are formed, we can add an organic solvent to each reaction mixture. Bromine dissolves and forms an orange organic layer while iodine dissolves and forms a purple organic layer.

CE 2002 Paper 2 Q6
ASL 1999 Q12(b)(iii)

53.10 Identifying six solutions from results of mixing pairs of solutions

Each of six reagent bottles labelled A, B, C, D, E and F contained one of the following solutions:

$AgNO_3(aq)$, $CuSO_4(aq)$, $HCl(aq)$, $NH_3(aq)$, $NaOH(aq)$, $Na_2S_2O_3(aq)$

In an attempt to identify the solutions in the bottles, series of tests were conducted by mixing pairs of the solutions. The table shown below lists the observations in these tests.

Solutions being mixed	Observations
A and C	A blue precipitate was formed.
A and F	A blue precipitate was formed initially, and the precipitate dissolved when F was in excess.
B and C	Only heat was liberated.
B and D	A pale yellow precipitate was formed slowly.
B and E	A white precipitate was formed.

Identify, with explanation and appropriate equations, the solutions in the six reagent bottles based on the observations.

N22 Examination questions often ask about identifying solutions of bromide ions / iodide ions by adding aqueous chlorine.

Solution

A was $CuSO_4(aq)$ as it reacted with C and F to give a blue precipitate ($Cu(OH)_2$):

$$Cu^{2+}(aq) + 2OH^-(aq) \longrightarrow Cu(OH)_2(s)$$

C and F are alkalis because $CuSO_4(aq)$ reacted with the alkalis to give a blue precipitate. As the blue precipitate dissolved when F was in excess, F was $NH_3(aq)$. The blue precipitate dissolved due to the formation of complex ions:

$$Cu(OH)_2(s) + 4NH_3(aq) \longrightarrow [Cu(NH_3)_4]^{2+}(aq) + 2OH^-(aq)$$

Thus, C was NaOH(aq).

B was HCl(aq) as it underwent neutralization with C and heat was liberated.

$$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$$

E was $AgNO_3(aq)$ as it reacted with $Cl^-(aq)$ ions (in B) to give a white precipitate (AgCl):

$$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$$

D was $Na_2S_2O_3(aq)$ as it reacted with an acid (B) to give a pale yellow precipitate (S):

$$S_2O_3^{2-}(aq) + 2H^+(aq) \longrightarrow S(s) + SO_2(g) + H_2O(l)$$

To summarize, the six solutions were:

A — $CuSO_4(aq)$

B — HCl(aq)

C — NaOH(aq)

D — $Na_2S_2O_3(aq)$

E — $AgNO_3(aq)$

F — $NH_3(aq)$

In the following activity, you are going to identify the anion present in an unknown solid sample.



Activity 53.2

Identifying the anion in an unknown solid sample.



Checkpoint

Refer to page T33.

1 For each of the following pairs of chemicals, suggest a chemical test to distinguish one from the other and state the expected observations.

a) Dilute hydrochloric acid and dilute sulphuric acid

b) Table salt and white sugar

c) Aqueous solutions of potassium chloride and potassium bromide

2 A colourless substance X contains one cation and one anion. The following table shows the observations of tests on an aqueous solution of X.

Test	Procedure	Observation
I	Add dilute nitric acid to an aqueous solution of X, followed by an aqueous solution of silver nitrate.	A white precipitate was formed.
II	Add dilute aqueous ammonia to an aqueous solution of X until excess.	A white precipitate was formed initially, and the precipitate was soluble in excess aqueous ammonia.

a) i) Name the anion in X.

ii) Write an ionic equation for the formation of the white precipitate in Test I.

b) i) Name the cation in X.

ii) Explain all the observations in Test II and write appropriate equations.

53.19 Devising a scheme to separate anions in an aqueous solution

Suppose we have an aqueous solution containing two or more anions. We can separate the anions by precipitation with barium ions and silver ions.

Carbonate ions, sulphite ions and sulphate ions form insoluble barium salts. Thus, we can precipitate these anions from a solution by adding aqueous solution of barium chloride or nitrate.

Chloride ions, bromide ions and iodide ions form insoluble silver salts. Thus, we can precipitate these anions from a solution by adding aqueous solution of silver nitrate.

Suppose we are given an aqueous solution containing three anions:

$Cl^-(aq)$, $I^-(aq)$, $SO_4^{2-}(aq)$

How can we separate the anions in the solution? We can classify the anions into two groups as follows:

- $SO_4^{2-}(aq)$ ions form an insoluble barium salt;
- $Cl^-(aq)$ ions and $I^-(aq)$ ions form insoluble silver salts.

Study the flow diagram shown in Fig. 53.29 to see how we can separate the anions. Notice the following points:

- precipitate of AgCl(s) obtained in *Step 2* is soluble in aqueous ammonia;
- AgCl(s) can be recovered by acidifying the solution containing $[Ag(NH_3)_2]^+(aq)$ and $Cl^-(aq)$ (see *Step 4*).

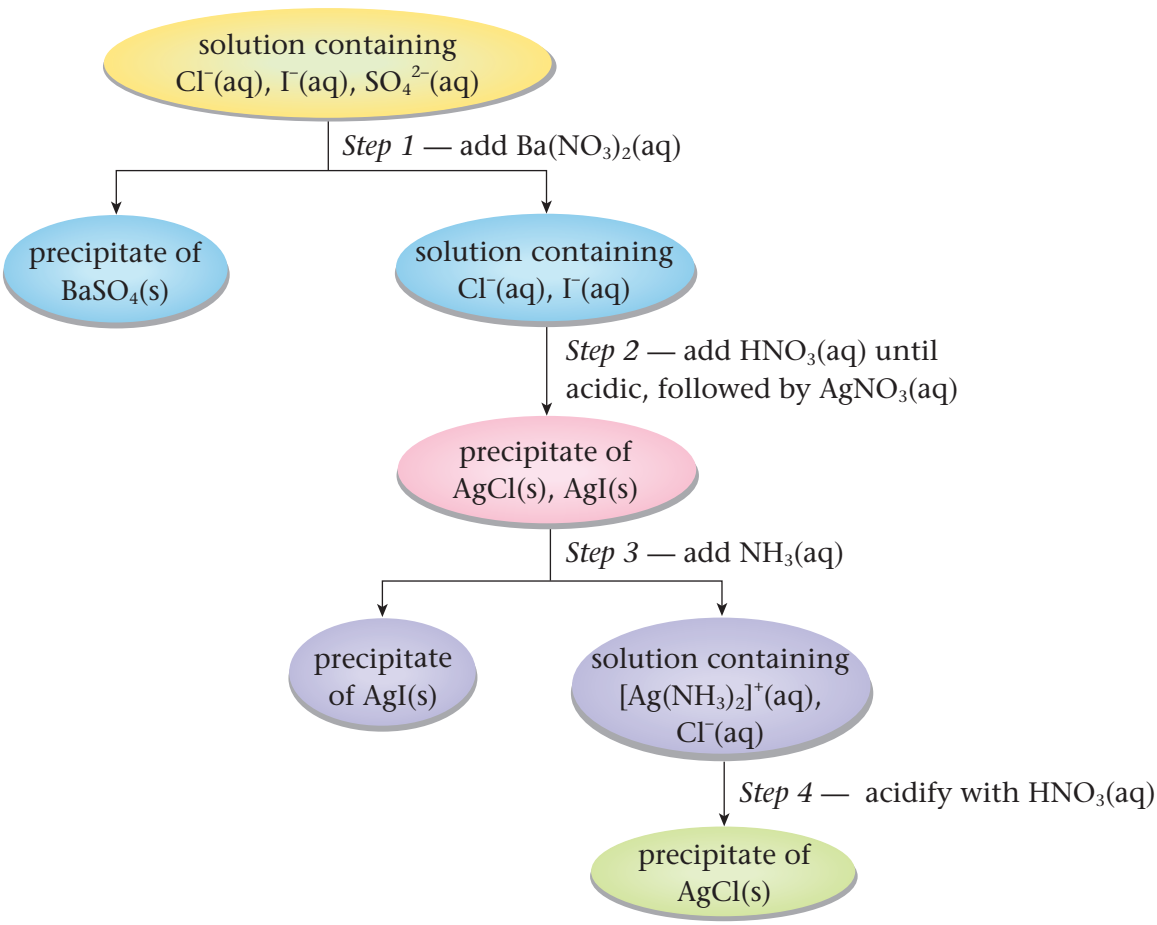
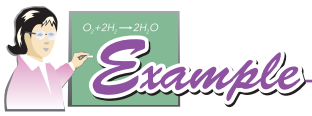


Fig. 53.29 A separation scheme for an aqueous solution containing three anions: $Cl^-(aq)$, $I^-(aq)$, $SO_4^{2-}(aq)$

53.20 Identifying the cation and anion in an unknown sample in the laboratory

After studying the action of various reagents on cations and anions, you are going to design and perform an investigation to deduce the cation and anion in a sample.



53.11 Identifying the cations and anion in a sample from observations of tests

Consider a solid sample X containing two cations and one anion. Some tests are carried out on X and the observations are shown in the table below.

a) Complete the last column on the deduction you can make based on the observation of each test.

b) Identify, with explanation, the cations and anion in X.

Solution

Test	Procedure	Observations	Deduction
1	Describe the appearance of X.	X is a pale violet crystalline solid.	X may contain transition metal ions.
2	Mix X with calcium hydroxide and heat gently. Test any gas given off with moist red litmus paper.	A brown solid is obtained on heating. A pungent gas is given off and it turns moist red litmus paper blue.	The gas given off is ammonia. Thus, X contains ammonium ions.
3	Prepare an aqueous solution of X. Divide the solution into two portions, one for each of the following tests.	A yellow-brown solution forms.	—
	a) Add dilute aqueous solution of sodium hydroxide.	A reddish brown precipitate forms.	X contains iron(III) ions.
	b) Add dilute hydrochloric acid, followed by aqueous solution of barium chloride.	A white precipitate forms.	The white precipitate is barium sulphate, thus showing that X contains sulphate ions.

From Test 1, we can deduce that X may contain transition metal ions as it is coloured.

From Test 2, we can deduce that X contains ammonium ions as ammonia gas is evolved when it is heated with calcium hydroxide.

From Test 3(a), we can deduce that X contains iron(III) ions as a reddish brown precipitate (iron(III) hydroxide) forms upon the action of dilute aqueous solution of sodium hydroxide.

From Test 3(b), we can deduce that X contains sulphate ions as a white precipitate (barium sulphate) is formed upon the action of dilute hydrochloric acid followed by aqueous solution of barium chloride.

Therefore we can conclude that sample X is ammonium iron(III) sulphate. Observations of the brown solid* obtained upon heating (Test 2) and the yellow-brown colour solution formed when X dissolves in water (Test 3) agree with the conclusion made.

* The brown solid is iron(III) oxide.

Complete the following problem to prepare yourself for the identification task in the laboratory.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Problem Solving

Consider a solid sample Y containing one cation and one anion. Some tests are carried out on Y and the observations shown in the table below.

a) Complete the last column on the deduction you can make based on the observation of each test.

b) Identify, with explanation, the cation and anion in Y.

Test	Procedure	Observations	Deduction
1	Describe the appearance of Y.	Y is a white crystalline solid.	Y does not contain coloured transition metal ions.
2	Carry out a flame test on Y.	A lilac flame appears.	Y contains potassium ions.

Continued on next page

Y contains potassium ions because Y gives a lilac flame in the flame test.
Y contains halide ions because the aqueous solution of Y gives a creamy white precipitate with acidified aqueous solution of silver nitrate.
The precipitate formed is silver bromide which does not dissolve in excess dilute aqueous ammonia but soluble in very concentrated aqueous ammonia.
Thus, Y contains bromide ions.

Test	Procedure	Observations	Deduction
3	Prepare an aqueous solution of Y.	A colourless solution forms.	—
	a) Add dilute nitric acid followed by aqueous solution of silver nitrate. Divide the reaction mixture into two separate portions.	A creamy white precipitate forms.	Y contains halide ions.
	b) To the first portion, add excess dilute aqueous ammonia.	The precipitate does not dissolve.	Y does not contain chloride ions.
	c) To the second portion, add excess concentrated aqueous ammonia.	The precipitate dissolves, giving a colourless solution.	Y contains bromide ions.

53.21 Risks associated with chemical tests

In the activity 53.3, you are going to design and carry out an investigation to identify four unlabelled solid samples. When designing your investigation, you need to assess the risk involved.

Table 53.10 gives the types of hazardous chemicals, their hazardous nature and general control measures and precautions to take when handling them.

* The general safety precautions here are only general guidelines. Refer to the precautions of each individual chemical as required.

Table 53.10 Nature of hazardous chemicals

Type of hazardous chemicals	Hazardous nature	General safety precautions*
Explosive 	substances that may explode if ignited in air or exposed to heat; a sudden shock or friction may start an explosion too	• keep away from heat and naked flames • avoid shock or friction

Continued on next page

Type of hazardous chemicals	Hazardous nature	General safety precautions*
Oxidizing  AL 2005 Paper 1 Q7(a)(ii)	substances that may produce a lot of heat when reacting with flammable or easily oxidizable substances; can create a fire risk	• keep away from heat and naked flames • wash affected areas with plenty of water if spilt on the skin or clothes; report the accident to the teacher
Flammable 	substances that may catch fire easily under normal conditions	• keep their reagent bottles tightly closed • keep away from heat and naked flames • keep in a well ventilated place
Carcinogenic 	substances that cause cancer or are suspected of causing cancer	• wear safety glasses, protective gloves and protective clothes • perform the experiment inside a fume cupboard • avoid breathing in their vapours
Toxic 	substances that may cause serious risk to health and even death if swallowed, breathed in or absorbed through the skin	• wear protective gloves • perform the experiment inside a fume cupboard • avoid breathing in their vapours • keep in a well ventilated place
Corrosive 	substances that damage living tissue upon contact, including the eyes and skin	• wear safety glasses and protective gloves • perform the experiment inside a fume cupboard • wash affected areas with plenty of water if spilt on the skin or clothes; report the accident to the teacher
Harmful 	harmful substances may involve limited health risk if swallowed, breathed in or absorbed through the skin	• avoid breathing in their vapours • wear protective gloves • wash affected areas with plenty of water if spilt on the skin or clothes; report the accident to the teacher
Irritant 	irritant substances are not corrosive but cause reddening or *blistering* of the skin	• avoid contact with the skin or clothing • wash affected areas with plenty of water if spilt on the skin or clothes; report the accident to the teacher

blistering 起水泡

Risk assessment

Follow the steps listed below when making the risk assessment.

- Write down the procedure of the tests you plan to use (chemicals involved, amounts, concentrations, techniques and equipment).
- Identify the hazards involved (hazardous chemicals and operations).
- Assess how likely the hazards will actually cause harm (risk).
- Decide what safety precautions to take, e.g. working inside the fume cupboard to avoid the spread of harmful gases.
- Find out how to dispose of any hazardous chemicals properly.

Suppose you plan to conduct a flame test with the steps* below:

1 Light a Bunsen burner. Half open the air hole of the burner to obtain a blue flame.

2 Place 10 cm^3 of concentrated hydrochloric acid in an evaporating dish.

3 Clean the end of a nichrome wire by dipping it into concentrated hydrochloric acid and then heat it in the Bunsen flame.

4 Heat and observe the colour of the flame. If the flame is still coloured, repeat Step 3.

Table 53.11 shows the risk assessment of the experiment.

* Only the first few steps of the flame test are listed here.

Table 53.11 Risk assessment of flame test

Hazardous chemical, procedure or equipment involved	Nature of hazard	Safety precautions	Source of information*
Bunsen burner	fire	• wear safety glasses • put the Bunsen burner on a fireproof mat • turn off the Bunsen burner when it is not in use	previous instruction

Continued on next page

Hazardous chemical, procedure or equipment involved	Nature of hazard	Safety precautions	Source of information*
Concentrated hydrochloric acid	corrosive	• wear safety glassses and protective gloves • carry out this activity inside a fume cupboard as concentrated hydrochloric acid gives off harmful fumes • wash affected areas with plenty of water if spilt on the skin or clothes; report the accident to the teacher • wash the container of concentrated hydrochloric acid with great care; do not add water to the acid directly • dispose of the acid in a waste bottle inside the fume cupboard	Material Safety Data Sheets (MSDS)
Nichrome wire	the glass portion may break if heated strongly	• do not heat the glass portion of the nichrome wire	previous instruction

* You may refer to the following information sources:
- Search for Material Safety Data Sheets (MSDS) from the website of City University of Hong Kong http://www.cityu.edu.hk/cse/safety
- Material Safety Data Sheets (MSDS) http://www.ilpi.com/msds/#Government
- International Chemical Safety Cards (ICSCs) http://www.ilo.org/dyn/icsc/showcard.home



Checkpoint

Read the procedure of the following experiment carefully. Identify all the hazards in the experiment. Find the relevant safety information and present the risk assessment in table form as shown below.

Experiment: Investigating the chemical properties of hex-1-ene

1 a) Place 1 cm^3 of hex-1-ene in a test tube.

b) Add a few drops of aqueous bromine to the test tube. Stopper and shake. Record the observations.

2 a) Place 1 cm^3 of hex-1-ene in a test tube.

b) Add a few drops of cold acidified dilute aqueous solution of potassium permanganate to the test tube. Stopper and shake. Record the observations.

Experiment title:			
Hazardous chemical, procedure or equipment involved	**Nature of hazard**	**Safety precautions**	**Source of information**

In the following activity, you are going to identify four unlabelled white solid samples.

Assessment Form

An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

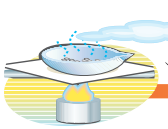


Activity 53.3

Identifying four unlabelled white solid samples.

Checkpoint

Experiment title: Investigating the chemical properties of hex-1-ene			
Hazardous chemical, procedure or equipment involved	**Nature of hazard**	**Safety precautions**	**Source of information**
Hex-1-ene	flammable, irritant	• wear safety glasses and protective gloves • keep the reagent bottle tightly closed • ensure good ventilation in the laboratory • dispose of hex-1-ene in a waste bottle inside the fume cupboard	Material Safety Data Sheets (MSDS)
Aqueous bromine	harmful	• keep aqueous bromine inside the fume cupboard • wear safety glasses and potective gloves • carry out the experiment inside the fume cupboard • wash affected areas with plenty of water if spilt on the skin or clothes; report the accident to the teacher • dispose of aqueous bromine in a waste bottle inside the fume cupboard	Material Safety Data Sheets (MSDS)
Shaking the solution mixtures	spillage may occur if shaken too vigorously	• do not shake too vigorously	previous instruction



Summary

1 Analytical chemistry consists of qualitative and quantitative analyses.

2 We can identify the cation in a sample using the following tests:

a) appearance;

b) solubility in water;

c) flame test;

d) action of heat;

e) action of dilute hydrochloric acid;

f) action of dilute sulphuric acid;

g) action of dilute aqueous solution of sodium hydroxide;

h) action of dilute aqueous ammonia.

3 The following table summarizes results of some tests for cations.

Test / Ion	Action of NaOH(aq)	Action of NH_3(aq)	Other test(s)
Aluminium ion, Al^{3+}	white precipitate, soluble in excess NaOH(aq) to give a colourless solution	white precipitate	—
Ammonium ion, NH_4^+	colourless gas (NH_3) given off on warming	—	—
Calcium ion, Ca^{2+}	white precipitate	—	flame test — brick-red flame action of H_2SO_4(aq) — white precipitate
Copper(II) ion, Cu^{2+}	blue precipitate	blue precipitate, soluble in excess NH_3(aq) to give a deep blue solution	flame test — bluish green flame
Iron(II) ion, Fe^{2+}	green precipitate, turning brown on standing	green precipitate, turning brown on standing	—

Continued on next page

Ion \ Test	Action of NaOH(aq)	Action of NH_3(aq)	Other test(s)
Iron(III) ion, Fe^{3+}	reddish brown precipitate	reddish brown precipitate	—
Lead(II) ion, Pb^{2+}	white precipitate, soluble in excess NaOH(aq) to give a colourless solution	white precipitate	action of HCl(aq) — white precipitate, soluble in hot water
			action of H_2SO_4(aq) — white precipitate
Magnesium ion, Mg^{2+}	white precipitate	white precipitate	—
Potassium ion, K^+	—	—	flame test — lilac flame
Sodium ion, Na^+	—	—	flame test — golden yellow flame
Zinc ion, Zn^{2+}	white precipitate, soluble in excess NaOH(aq) to give a colourless solution	white precipitate, soluble in excess NH_3(aq) to give a colourless solution	—

4 We can identify the anion in a sample using the following tests:

a) action of heat;

b) action of dilute hydrochloric acid;

c) action of concentrated sulphuric acid;

d) action of aqueous solution of silver nitrate.

5 The following table summarizes results of some tests for anions.

Ion \ Test	Action of dilute HCl(aq) or H_2SO_4(aq)	Action of $BaCl_2$(aq)	Action of concentrated H_2SO_4(aq)	Action of dilute HNO_3(aq), followed by $AgNO_3$(aq)	Other reagent(s)
Carbonate ion, CO_3^{2-}	colourless gas (CO_2) given off	white precipitate soluble in dilute HCl(aq) with colourless gas (CO_2) given off	vigorous reaction, colourless gas (CO_2) given off	—	—

Continued on next page

Ion \ Test	Action of dilute HCl(aq) or H_2SO_4(aq)	Action of $BaCl_2$(aq)	Action of concentrated H_2SO_4(aq)	Action of dilute HNO_3(aq), followed by $AgNO_3$(aq)	Other reagent(s)
Hypochlorite ion, OCl^-	greenish yellow gas (Cl_2) given off	—	—	—	—
Sulphite ion, SO_3^{2-}	colourless gas (SO_2) given off	white precipitate soluble in dilute HCl(aq) with colourless gas (SO_2) given off	vigorous reaction, colourless gas (SO_2) given off	—	—
Sulphate ion, SO_4^{2-}	—	white precipitate insoluble in dilute HCl(aq)	—	—	—
Chloride ion, Cl^-	—	—	steamy fumes (HCl) given off	white precipitate soluble in NH_3(aq)	—
Bromide ion, Br^-	—	—	reddish brown vapour (HBr, Br_2 and SO_2) given off	creamy precipitate soluble in concentrated NH_3(aq)	action of Cl_2(aq) + organic solvent — an orange organic layer forms*
Iodide ion, I^-	—	—	purple vapour (I_2, SO_2 and H_2S) given off	yellow precipitate	action of Cl_2(aq) + organic solvent — a purple organic layer forms*

* Chlorine oxidizes bromide ions to bromine.
$Cl_2(aq) + 2Br^-(aq) \longrightarrow 2Cl^-(aq) + Br_2(aq)$
The bromine dissolves in the organic solvent to form an orange organic layer.

* Chlorine oxidizes iodide ions to iodine.
$Cl_2(aq) + 2I^-(aq) \longrightarrow 2Cl^-(aq) + I_2(aq)$
The iodine dissolves in the organic solvent to form a purple organic layer.

6 Risk assessment is important in all chemical tests.



Exercise

Note: The symbol 🌶 indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 a) Describe the procedure of flame test.
Refer to page T34.
b) Describe how to distinguish a calcium salt and a sodium salt of the same anion by flame test.

2 Complete the table on the results of tests on some cations.
Refer to page T34.

Cation	Action of NaOH(aq)	Action of NH_3(aq)
Aluminium ion, Al^{3+}(aq)		
Ammonium ion, NH_4^+(aq)		
Calcium ion, Ca^{2+}(aq)		
Copper(II) ion, Cu^{2+}(aq)		
Iron(II) ion, Fe^{2+}(aq)		
Iron(III) ion, Fe^{3+}(aq)		
Lead(II) ion, Pb^{2+}(aq)		
Zinc ion, Zn^{2+}(aq)		

3 Complete the following table on the tests for some common gases and vapours.
Refer to page T35.

Gas / vapour	Test	Observation
Ammonia		
Carbon dioxide		
Chlorine		
Bromine		
Iodine		
Hydrogen		
Oxygen		
Hydrogen halides		
Sulphur dioxide		
Water vapour		

4 Complete the following table on the results of tests on some anions.
Refer to page T35.

Anion	Action of HCl(aq)	Action of concentrated H_2SO_4(aq)	Action of HCl(aq), followed by $BaCl_2$(aq)	Action of HNO_3(aq), followed by $AgNO_3$(aq)
Carbonate ion, CO_3^{2-}				—
Hypochlorite ion, OCl^-		—	—	—
Sulphite ion, SO_3^{2-}				—
Sulphate ion, SO_4^{2-}	—	—		—
Chloride ion, Cl^-	—		—	
Bromide ion, Br^-	—		—	
Iodide ion, I^-	—		—	

5 a) Eight hazard warning labels are shown below.


Explosive


Flammable


Toxic


Carcinogenic


Oxidizing


Corrosive



Harmful



Irritant

Write down the label(s) that should be displayed on a bottle containing each of the following chemicals:

i) $CH_3OH(l)$
Flammable and toxic

ii) $NaOH(s)$
Corrosive

iii) $KIO_3(s)$
Oxidizing

iv) Concentrated H_2SO_4
Corrosive

b) List the precautions you would take when working with each of the following chemicals:

i) $CH_3OH(l)$
- As methanol is flammable, keep the reagent bottle tightly closed.
- There should be no naked flame in the laboratory during an experiment with methanol.

ii) Concentrated H_2SO_4
- Wear safety glasses and protective gloves.
- Concentrated sulphuric acid is highly corrosive. It gives severe burns to the skin. Handle with great care.
- NEVER add water to concentrated sulphuric acid.
- If you spill any acid on your skin or clothes, wash it off immediately with plenty of water. Then report the accident to your teacher.
- Wash containers of concentrated sulphuric acid with great care; do not add water to the acid directly.
- Dispose of the acid in a waste bottle inside the fume cupboard.

Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

6 For each of the following experiments, state the expected observations.

a) Dilute aqueous ammonia is added dropwise, until in excess, to an aqueous solution of copper(II) sulphate. A pale blue precipitate forms initially.
The precipitate dissolves in excess dilute aqueous ammonia to form a deep blue solution.

b) Solid potassium bromide is heated with concentrated sulphuric acid. Reddish brown vapour is formed.
The vapour condenses to a reddish brown liquid.

c) Concentrated sulphuric acid is added to potassium iodide solid.
Purple vapour which condenses to a black solid is formed.

d) Steam is passed over a piece of anhydrous cobalt(II) chloride paper.
Anhydrous cobalt(II) chloride paper turns from blue to pink.

7 Consider the following pairs of solutions:
Refer to page T36.

a) $AgNO_3(aq) + HCl(aq)$

b) $Ca(NO_3)_2(aq) + MgSO_4(aq)$

c) $KCl(aq) + NH_3(aq)$

d) $ZnSO_4(aq) + K_2CO_3(aq)$

i) For each pair, decide whether the solutions will produce a precipitate when mixed.

ii) For each case in which a precipitate is produced, state the colour of the precipitate and write the relevant equation.

8 Answer each of the following questions by choosing one of the solids listed below. Give an appropriate equation in each case.
Refer to page T36.

$CuCO_3$, $FeSO_4 \bullet 7H_2O$, NaOCl, $Na_2S_2O_3$, KCl, KI

a) Which solid gives a black solid upon heating?

b) Which solid gives a colourless vapour that turns blue cobalt(II) chloride paper pink upon heating?

c) Which solid gives a gas that turns acidified aqueous solution of potassium dichromate green upon reaction with dilute hydrochloric acid?

d) Which solid gives a gas that turns limewater milky upon reaction with dilute hydrochloric acid?

e) Which solid gives a gas that bleaches moist blue litmus paper upon reaction with dilute hydrochloric acid?

f) Which solid gives a purple vapour upon reaction with concentrated sulphuric acid?

9 Briefly describe how you would show the presence of the following species in a sample of the salt, $CuSO_4 \bullet (NH_4)_2SO_4 \bullet 6H_2O$.

a) Copper(II) ion Add a few drops of dilute aqueous solution of sodium hydroxide to an aqueous solution of the salt. A pale blue precipitate forms.

b) Ammonium ion Warm the salt with dilute aqueous sodium hydroxide solution. Ammonia gas that turns moist red litmus paper blue is liberated.

c) Sulphate ion Add an aqueous solution of barium chloride followed by dilute hydrochloric acid to an aqueous solution of the salt. A white precipitate (barium sulphate) forms.

d) Water of crystallization Heat the salt in a boiling tube. The water vapour evolved turns blue cobalt(II) chloride paper pink.

10 For each of the following pairs of solid samples, suggest a test to distinguish between them. State the expected observations and write an equation for each reaction that takes place.
Refer to page T37.

a) Ammonium sulphate and sodium sulphate

b) Potassium nitrate and sodium nitrate

c) Copper(II) chloride and iron(II) sulphate

d) Magnesium chloride and magnesium carbonate

e) Sodium bromide and sodium iodide

f) Potassium sulphate and potassium sulphite

11 An aqueous solution of silver nitrate can be used as a test for halide ions. A student decided to carry out this test on a solution of magnesium chloride. The bottle of magnesium chloride that the student used showed the formula $MgCl_2 \bullet 6H_2O$.

a) The student dissolved a small amount of $MgCl_2 \bullet 6H_2O$ in water and added an aqueous solution of silver nitrate to the aqueous solution.

i) What would the student see after adding the aqueous solution of silver nitrate, $AgNO_3(aq)$?
A white precipitate forms.

ii) Write an ionic equation for this reaction. Include state symbols.
$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$

iii) Using an aqueous solution of silver nitrate, it is sometimes difficult to distinguish between chloride, bromide and iodide ions. AgCl(s) is readily soluble in aqueous ammonia.
AgBr(s) is soluble in very concentrated aqueous ammonia.
AgI(s) is insoluble in aqueous ammonia.
How can aqueous ammonia be used to distinguish between these three ions?

b) Domestic tap water has been chlorinated. Chlorine reacts with water as shown below.

$$Cl_2(g) + H_2O(l) \longrightarrow HOCl(aq) + HCl(aq)$$

i) State the oxidation number of chlorine in Cl_2, HOCl and HCl.
Cl_2 0
HOCl +1
HCl −1

ii) When carrying out halide tests with an aqueous solution of silver nitrate, it is important that distilled or deionized water is used for all solutions, rather than tap water. Suggest why.
Tap water contains chloride ions.

(OCR Advanced Subsidiary GCE, Foundation Chem, Jun. 2006)

12 You are provided with three unlabelled bottles each containing one of the white powders listed below:

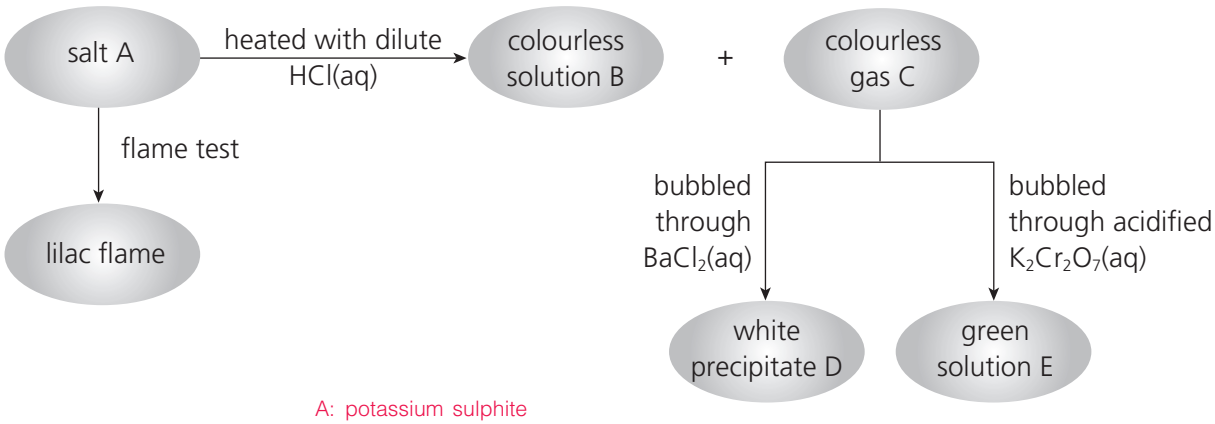
KBr(s), SiO_2(s) and glucose

a) Outline the *physical* tests that you would perform to distinguish unambiguously the three substances from one another.

b) Describe how you would carry out a *chemical* test to distinguish KBr(s) from glucose.

(HKALE 2004)

13 The following diagram shows some properties and reactions involving A, B, C, D and E. A is a normal salt.



a) Name A, B, C and D.
A: potassium sulphite
B: potassium chloride
C: sulphur dioxide
D: barium sulphite

b) Write an equation for the reaction between

i) A and dilute HCl(aq);
$K_2SO_3(aq) + 2HCl(aq) \longrightarrow 2KCl(aq) + H_2O(l) + SO_2(g)$

ii) C and $BaCl_2$(aq);
$Ba^{2+}(aq) + SO_3^{2-}(aq) \longrightarrow BaSO_3(s)$

iii) C and acidified $K_2Cr_2O_7$(aq).
$3SO_2(aq) + Cr_2O_7^{2-}(aq) + 2H^+(aq) \longrightarrow 3SO_4^{2-}(aq) + 2Cr^{3+}(aq) + H_2O(l)$

c) What would be observed when D is warmed with dilute hydrochloric acid?
A colourless solution would form.

d) Suggest another reagent that will give a white precipitate with an aqueous solution of barium chloride.
Any soluble carbonate / sulphate

e) In another experiment, gas C is bubbled into an aqueous solution of sodium hydroxide.

i) Name the type of reaction that takes place.
Neutralization

ii) Write a chemical equation for the reaction.
$SO_2(g) + 2NaOH(aq) \longrightarrow Na_2SO_3(aq) + H_2O(l)$

f) Draw a hazard warning label that should be displayed on a bottle of 2 mol dm^{-3} hydrochloric acid.



Exercise

14 a) Salt A contains one cation and one anion.

When a sample of A is heated with an aqueous solution of sodium hydroxide a gas, B, is given off that turns moist red litmus paper blue.

When dilute hydrochloric acid followed by an aqueous solution of barium chloride is added to a solution of A, a white precipitate, C, is formed.

Identify A, B and C.
A — ammonium sulphate
B — ammonia
C — barium sulphate

b) Salt D contains one cation and one anion.

When a flame test is carried out on D, a lilac colour is observed in the flame.

When concentrated sulphuric acid is added to D, brown fumes of an element, E, are given off together with a colourless gas, F.

F turns an acidified potassium dichromate solution green.

Identify D, E and F.
D — potassium bromide
E — bromine
F — sulphur dioxide

(Edexcel Advanced Subsidiary GCE, Unit Test 3B, Jun. 2007)

Exercise

15 A, B, C and D are four unlabelled bottles, each containing the aqueous nitrate solution of one of the following cations:

calcium ion, iron(II) ion, lead(II) ion, zinc ion

The results of the tests on the solutions are as follows:

Test		Bottle			
		A	**B**	**C**	**D**
I	Adding dilute aqueous solution of sodium hydroxide	green precipitate, turning brown on standing	white precipitate, soluble in excess NaOH(aq)	white precipitate, soluble in excess NaOH(aq)	white precipitate
II	Adding dilute aqueous solution of sodium carbonate	?	white precipitate	white precipitate	white precipitate

a) Consider the reagent in bottle A.

i) Identify the cation present.
Iron(II) ion

ii) Explain why the green precipitate formed in *Test I* turns brown on standing.
Iron(II) hydroxide is oxidized to iron(III) hydroxide on standing in air.

iii) State the expected observations in *Test II*.
A green precipitate ($FeCO_3(s)$) forms.

b) Consider the reagent in bottle D.

i) Identify the cation present.

Calcium ion

ii) Write an ionic equation for the formation of the white precipitate in *Test I*.

$Ca^{2+}(aq) + 2OH^{-}(aq) \longrightarrow Ca(OH)_2(s)$

c) Add dilute aqueous ammonia to each solution.
Zinc ion gives a white precipitate that is soluble in excess dilute aqueous ammonia.
Lead(II) ion gives a white precipitate that is insoluble in excess dilute aqueous ammonia.

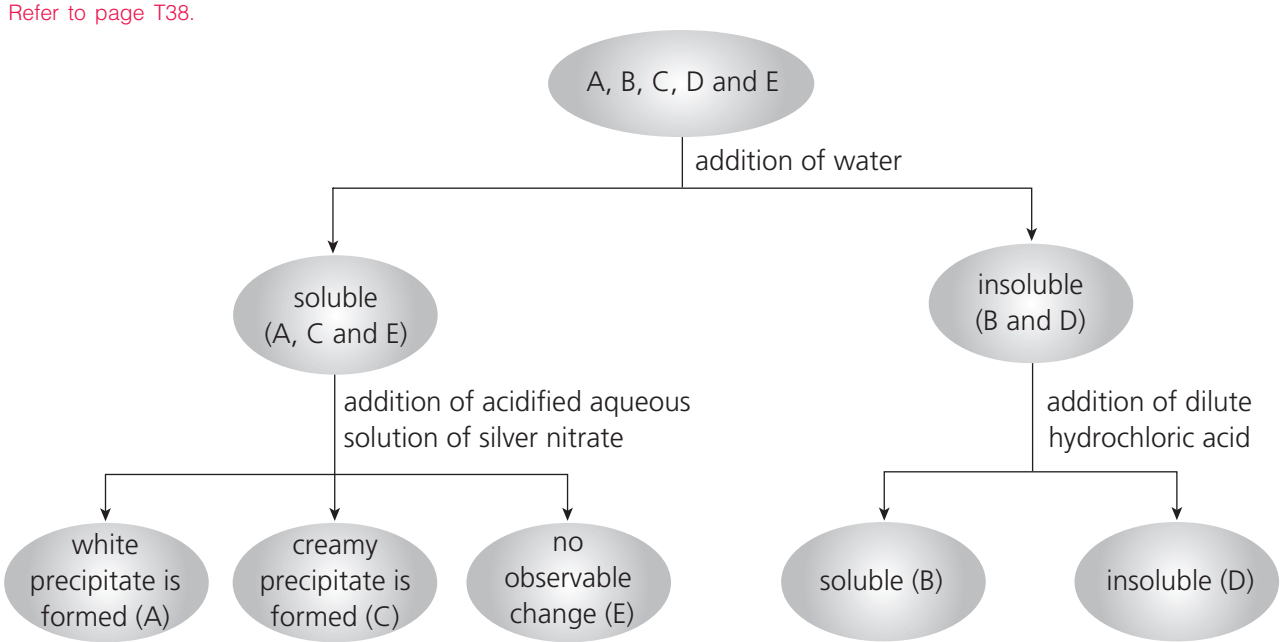
c) Cations present in bottles B and C cannot be distinguished easily using the above tests. Suggest another chemical test to distinguish them.

d) Draw a hazard warning label that should be displayed on a bottle of 2 mol dm^{-3} solution of sodium hydroxide.



16 Compounds A, B, C, D and E are ionic compounds. The following flow diagram is used to identify each compound.

Refer to page T38.



a) D gives a brick-red flame in flame test. Suggest what D might be. Explain your answer briefly.

b) B dissolves in dilute hydrochloric acid to give a blue solution and no other change is observed.

i) What might B be? Explain your answer.

ii) Write an ionic equation for the reaction between B and dilute hydrochloric acid.

c) When A is heated with a concentrated aqueous solution of sodium hydroxide, a gas which turns moist red litmus paper blue is evolved.

i) What can you conclude from this observation? Write an ionic equation for the reaction between concentrated aqueous solution of sodium hydroxide and A.

ii) Suggest the anion in A. Explain your answer with the help of a chemical equation.

d) i) Suggest the anion in C.

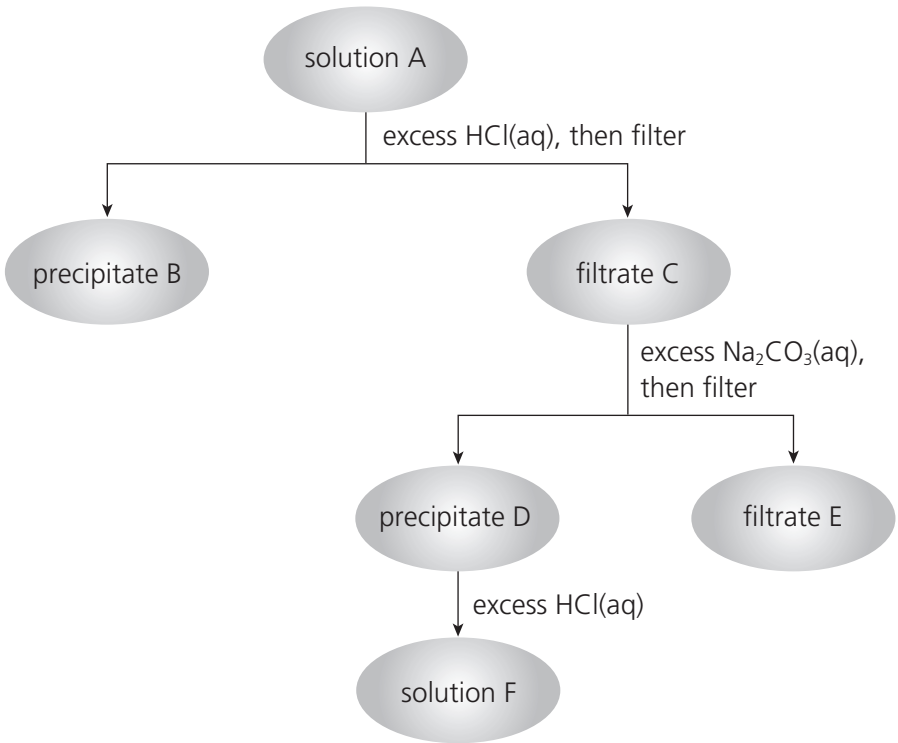
ii) What will be observed if chlorine gas is bubbled into an aqueous solution of C? Explain your answer with the help of an ionic equation.

e) Suppose E is a sodium compound, suggest a test to show the presence of sodium ions in it. State the expected observation.

17 A is a nitrate solution containing the following cations:

$Cu^{2+}(aq)$, $Mg^{2+}(aq)$, $Na^{+}(aq)$, $Pb^{2+}(aq)$

The following flow diagram shows an attempt to separate the above cations.



a) What is the colour of solution A?
Blue

b) i) Name B.
Lead(II) chloride

ii) Write an ionic equation for the formation of B from A.
$Pb^{2+}(aq) + 2Cl^{-}(aq) \longrightarrow PbCl_2(s)$

c) Write an ionic equation for the reaction of D with HCl(aq).
$CuCO_3(s)$ / $MgCO_3(s) + 2H^{+}(aq) \longrightarrow Cu^{2+}(aq)$ / $Mg^{2+}(aq) + H_2O(l) + CO_2(g)$

d) What would be observed if $NH_3(aq)$ is added dropwise to solution F until in excess? Write appropriate equations to account for your answer.
A pale blue precipitate forms.
$Mg^{2+}(aq) + OH^{-}(aq) \longrightarrow Mg(OH)_2(s)$
white precipitate
$Cu^{2+}(aq) + OH^{-}(aq) \longrightarrow Cu(OH)_2(s)$
pale blue precipitate
Copper(II) hydroxide dissolves in excess $NH_3(aq)$ to give a deep blue solution.
The white precipitate remains.
$Cu(OH)_2(s) + 4NH_3(aq) \longrightarrow [Cu(NH_3)_4]^{2+}(aq) + 2OH^{-}(aq)$

e) Which cation is present in E?
Sodium ion

18 Each of the six reagent bottles labelled A, B, C, D, E and F contained one of the following solutions:
Refer to page T38.

$BaCl_2(aq)$, $H_2SO_4(aq)$, $Fe(NO_3)_2(aq)$, $NaOH(aq)$, $NaOCl(aq)$ and $Na_2SO_3(aq)$

In an attempt to identify the contents of the bottles, a series of tests was conducted by mixing two of the solutions. The table below lists the observations in these tests.

Solutions being mixed	Observations
A and B	a gas is evolved; the gas turns a filter paper soaked with aqueous solution of potassium dichromate green
A and C	only heat is liberated
A and D	a white precipitate is formed
A and F	a gas is evolved; the gas turns moist blue litmus paper red and then bleaches it
C and E	a green precipitate is formed

Identify, with explanation and appropriate equations, the solutions in the six reagent bottles based on the observations.

19 Given a sample solution containing three anions:
Refer to page T39.

$Br^-(aq)$, $CO_3^{2-}(aq)$, $SO_4^{2-}(aq)$

Devise a separation scheme for the anions in the sample solution. Explain the rationale of your scheme and present it in the form of a flow diagram.

20 Suppose you are given six unlabelled reagent bottles, each containing one of the following solutions:
Refer to page T40.

$AgNO_3(aq)$, $HCl(aq)$, $Na_2SO_4(aq)$, $NH_3(aq)$, $ZnSO_4(aq)$, $ZnCl_2(aq)$

Given some test tubes but nothing else, describe briefly how you can label the bottles correctly. Record your answers in a table form as follows:

Test	Observation	Deduction

21 There are five unlabelled reagent bottles each containing one of the white solids listed below:
Refer to page T40.

ammonium nitrate, ammonium sulphate, sodium carbonate, sodium hypochlorite, sodium chloride

Suggest how you would carry out tests to distinguish the five solids from each other.

(For this question, you are required to give answers in paragraph form. Use equations, diagrams and examples where appropriate.)